

Department of Computer Science and Engineering (CSE)
BRAC University

Lecture 11

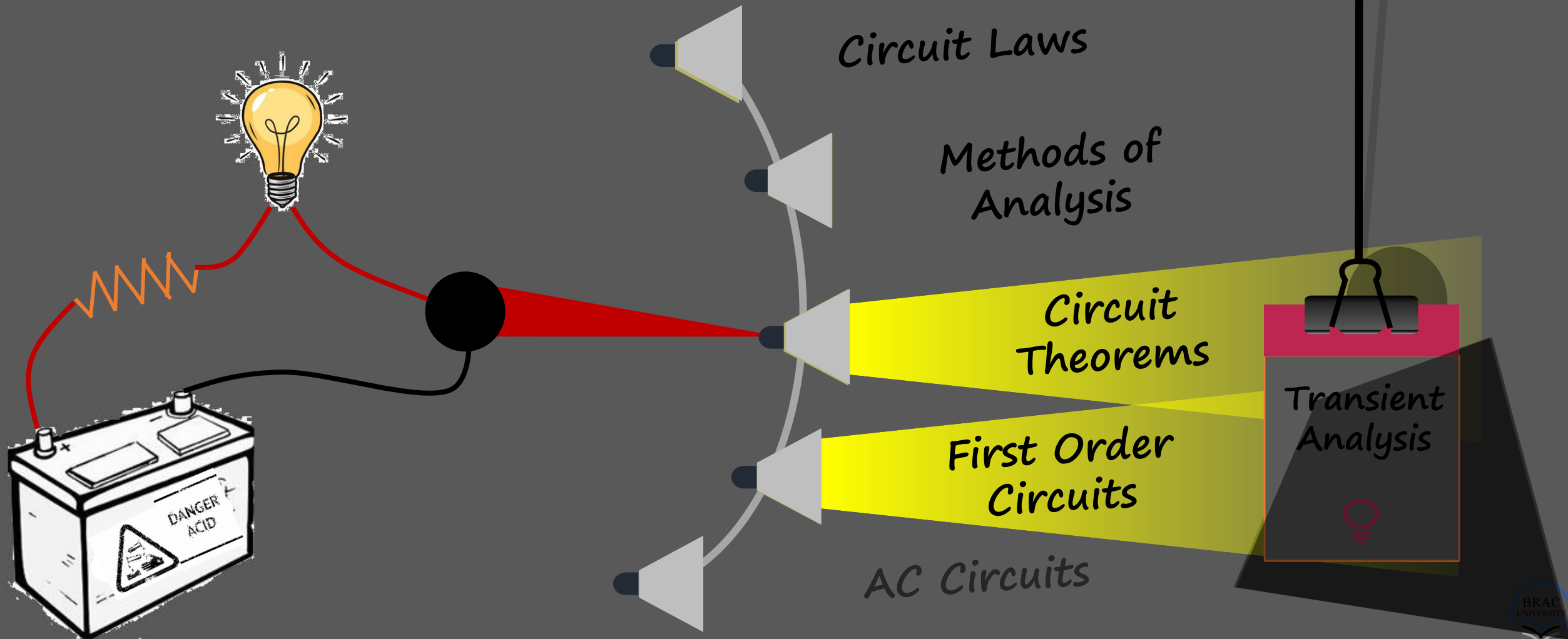
CSE250 - Circuits and Electronics

FIRST ORDER CIRCUITS



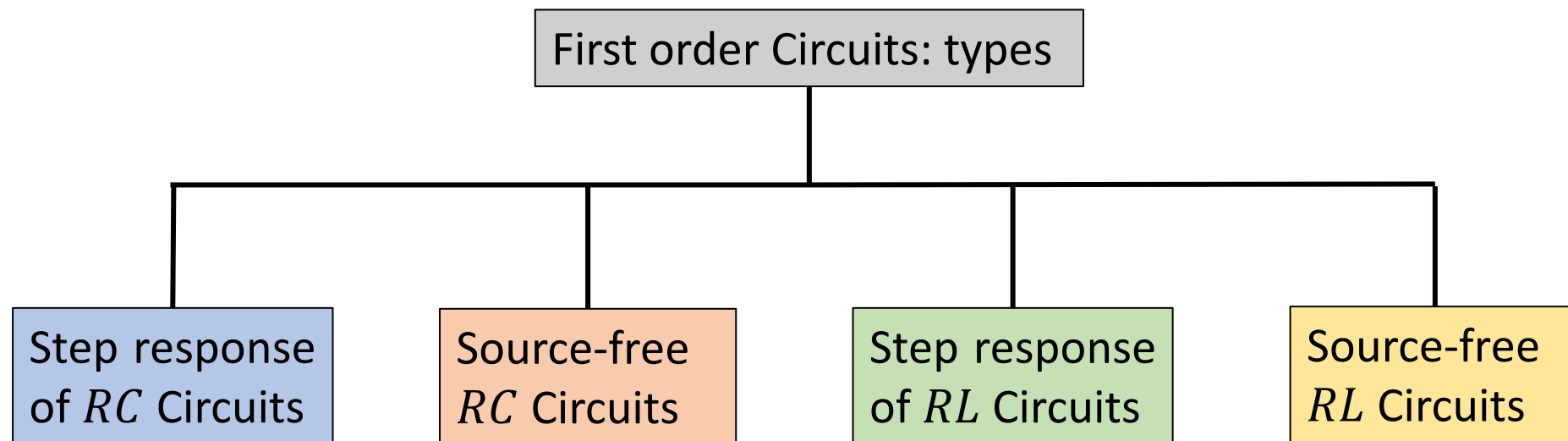
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Course Outline: broad themes



First Order Circuits

- A **first-order** circuit is characterized by a first-order differential equation.
- We shall examine two types of differential circuits: circuit comprising resistors and capacitors (RC circuit) and circuit comprising resistors and inductors (RL circuit).
- Two ways to excite the circuits: (i) by initial conditions of storage elements (source free circuits) and (ii) by independent sources (DC for this course).



Circuit Elements

- **Active element**

- An *active element* is capable of generating energy.
- In other words, an element is said to be active if it can add some gain (in terms of voltage or current) to a circuit.
- Active elements can absorb energy if they are forced to do so by other active elements.
- Examples: *Voltage/current sources, generators, transistors, operational amplifiers.*

- **Passive element**

- *Passive elements* cannot supply energy. They can only consume/dissipate/store energy.
- Examples: *Resistors, capacitors, inductors, transformers.*
- Transformers change the voltage or current levels, but the power is unchanged. This is why transformers are passive element.

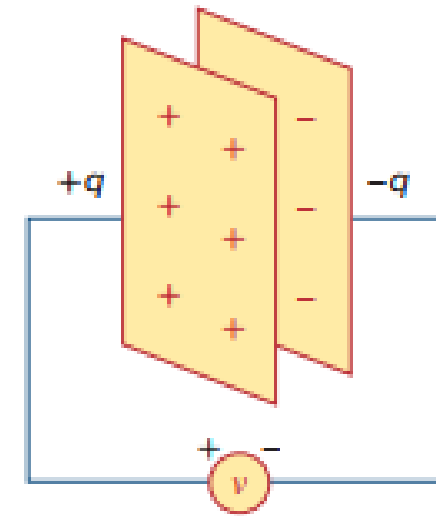
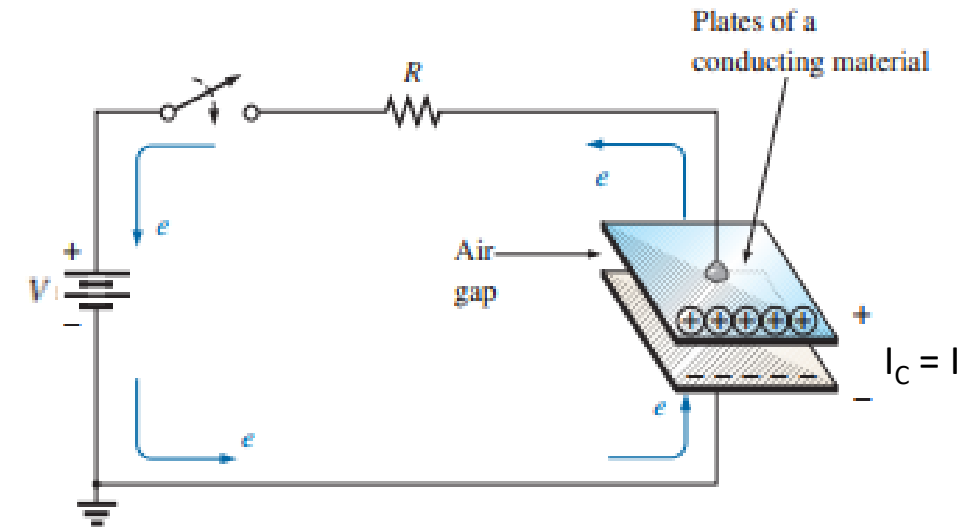
Capacitors

- A **capacitor** is a passive circuit element designed to store energy in its electric field.
- Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy, which can be retrieved at a later time. For this reason, capacitors and inductors are called *storage* elements.



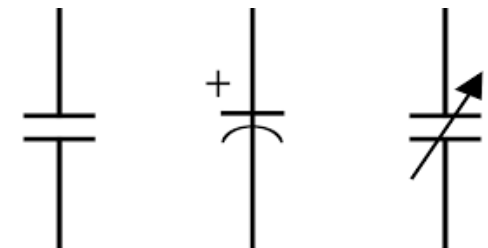
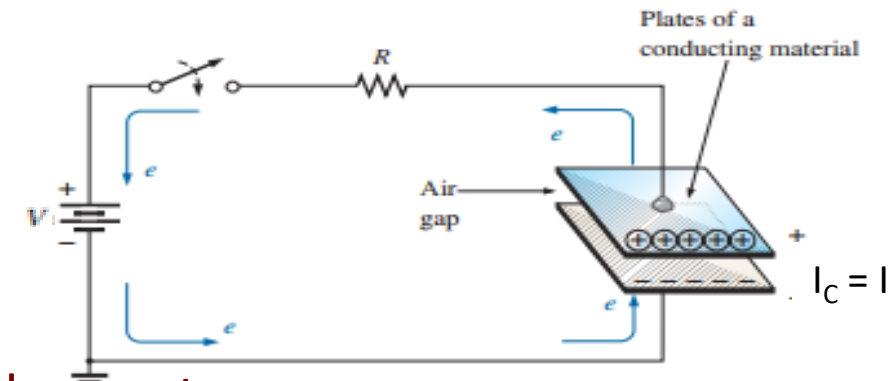
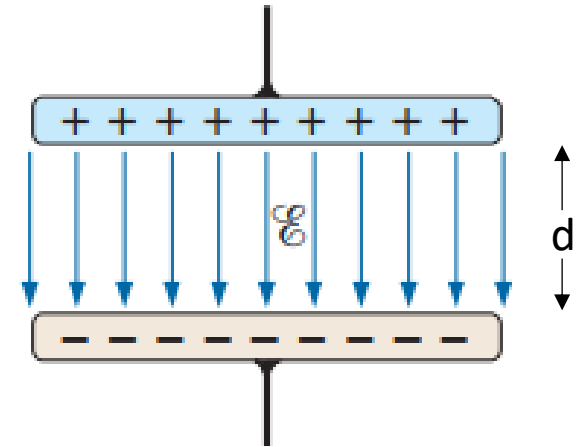
Parallel Plate Capacitor

- Most widely used configuration is the two conducting surfaces (aluminium mainly) separated by a dielectric (air, ceramic, paper, or mica).
- The switch is open initially (no net charge).
- Closing the switch causes electrons to flow from and to the upper and lower plates respectively as shown by the arrows.
- Electron flow continues until the potential difference between the plates equals the applied potential.
- The final result is a net positive charge on the top plate and a negative charge on the bottom plate.



Capacitance

- *Capacitance* is a measure of a capacitor's ability to store charge.
- Increasing V increases E as $E \propto \frac{V}{d}$ as long as d is constant. An increase in E field causes increased charge separation i.e. increases q .
- So, $q \propto V$
- $\Rightarrow q = CV$ [C is a proportionality constant $\equiv$ Capacitance]
- $\Rightarrow C = \frac{q}{V}$ [F (Farad), mF , μF]
- $\Rightarrow$ For a particular capacitor $\uparrow V, \uparrow q$ but $\frac{q}{V} = \text{const.}$ So, C does not depend on q or v . It depends on the physical dimension of the capacitor.
- $\Rightarrow$ For the parallel plate capacitor, $C = \frac{\epsilon A}{d}$



Fixed Capacitor Polarized Capacitor Variable Capacitor

I-V relation of a Capacitor

- From the definition of the capacitance,

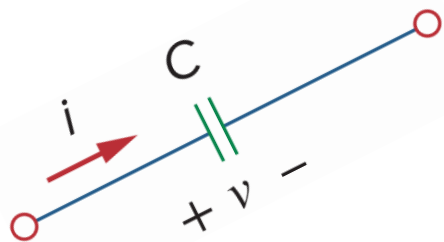
$$C = \frac{q}{v}$$

$$\Rightarrow q = Cv$$

- Differentiating with respect to time,

$$\frac{dq}{dt} = C \frac{dv}{dt}$$

$$i = C \frac{dv}{dt}$$



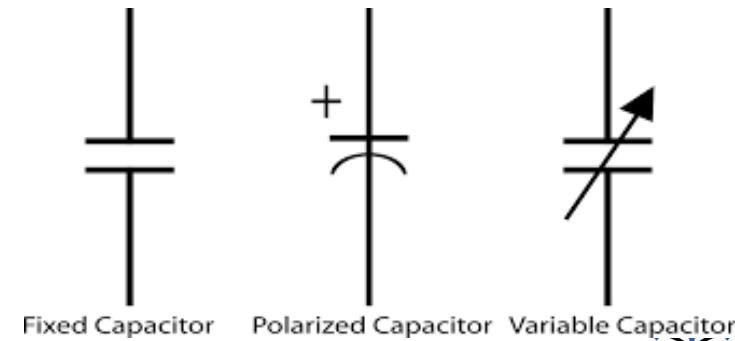
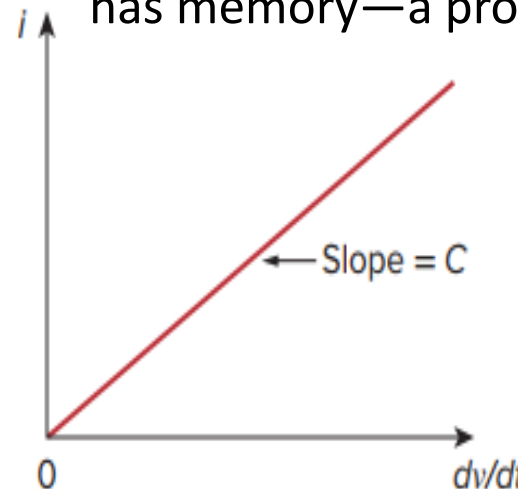
- This is the characteristic equation of a capacitor.
- Integrating with respect to time,

$$v(t) = \frac{1}{C} \int i(t) dt$$

- If the voltage of the capacitor at any time t_0 is $v(t_0) = q(t_0)/C$, then,

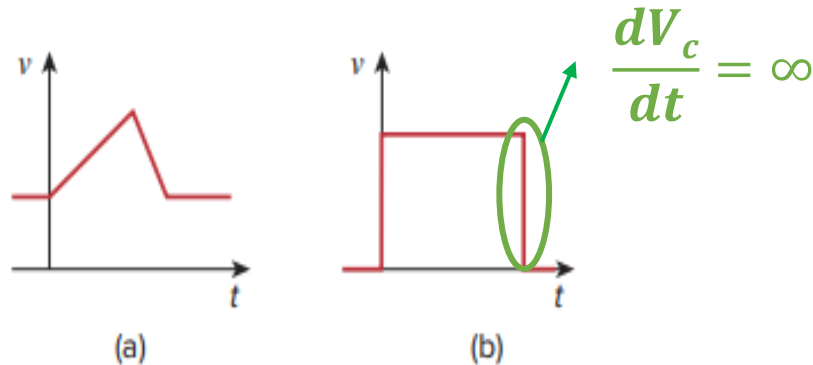
$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

- It shows that capacitor voltage depends on the past history of the capacitor current. Hence, the capacitor has memory—a property that is often exploited.



Capacitor: important properties

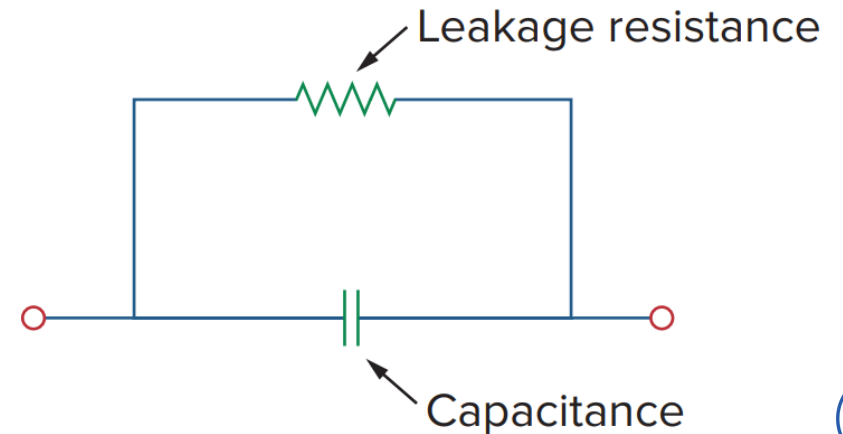
1. A capacitor is an open circuit to dc. At dc, $i_C = C \frac{dV_{c,dc}}{dt} = 0$ [Open circuit]
2. The voltage on a capacitor cannot change abruptly.



Voltage change across a capacitor
(a) allowed and (b) not allowed

3. An ideal capacitor does not dissipate energy.

4. A real, nonideal capacitor has a parallel-model leakage resistance.



Step Response of a RC circuit

- The *step response* of a circuit is its behaviour under the sudden application of dc voltage or current source. We assume the circuit response to be the capacitor voltage.

⇒ Since the voltage of a capacitor cannot change instantaneously,

$$\Rightarrow v(0^-) = v(0^+) = V_0 \quad \Rightarrow [\ln(v - V_s)]_{V_0}^{v(t)} = -\left[\frac{t}{RC}\right]_0^t$$

$$\Rightarrow \text{Using KCL (for } t > 0), \quad \Rightarrow \ln[v(t) - V_s] - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

$$\Rightarrow C \frac{dv}{dt} + \frac{v - V_s}{R} = 0$$

$$\Rightarrow \frac{dv}{dt} = -\frac{v - V_s}{RC}$$

$$\Rightarrow \frac{dv}{v - V_s} = -\frac{1}{RC} dt$$

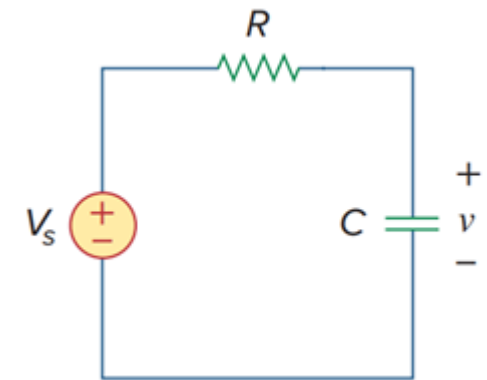
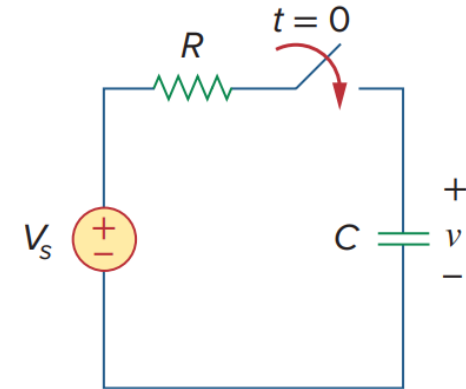
Integrating both sides,

$$\Rightarrow \ln \frac{v - V_s}{V_0 - V_s} = -\frac{t}{RC}$$

$$\Rightarrow \frac{v - V_s}{V_0 - V_s} = e^{-t/RC}$$

$$\Rightarrow v - V_s = (V_0 - V_s)e^{-t/RC}$$

$$\Rightarrow \mathbf{v(t) = V_s + (V_0 - V_s)e^{-t/RC}}$$



Time Constant

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{RC}}, & t > 0 \end{cases}$$

- This is known as the complete response (or total response) of the RC circuit to a sudden application of a dc voltage source. It is assumed that the capacitor was initially charged to V_0 .

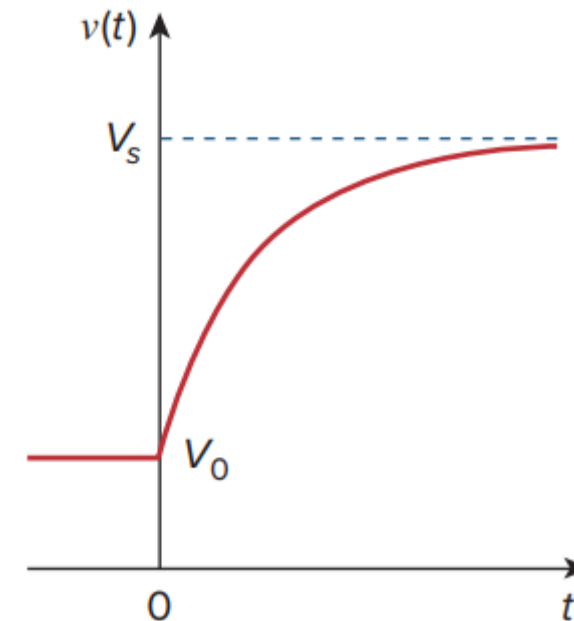
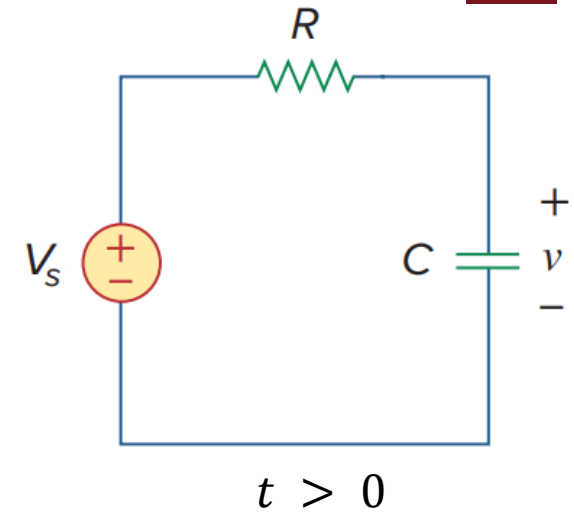
$$\Rightarrow v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$$

$\Rightarrow$ where $\tau = RC$ is the *time constant* (unit in sec).

No
Need

- Notice that, we write $\tau = RC$ for the circuit consisting of only a resistor R in series with the capacitor. As we know, all the linear two terminal circuits can be reduced to this form by Thevenin's Theorem, so the resistor R is actually the Thevenin Resistance R_{Th} . Therefore,

$$\tau = R_{Th}C$$



Transient and Steady-State Response

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$$

- The *complete response* can be broken into two parts—one temporary and the other permanent, that is,

$$v(t) = v_{ss} + v_t, \quad \text{where,}$$

$$v_{ss} = V_s \quad \& \quad v_t = (V_0 - V_s)e^{-\frac{t}{\tau}}$$

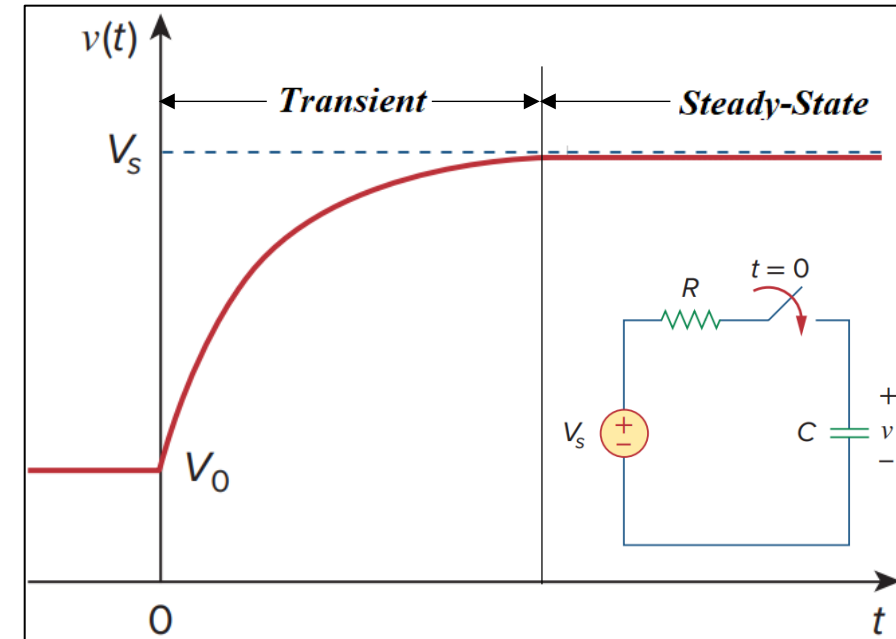
- The *transient response* (v_t) is the circuit's temporary response that will die out with time.

- The *steady-state response* (v_{ss}) is the behaviour of the circuit a long time after an external excitation is applied.

- The complete response can be written as,

$$v(t) = V_{final} + [V_{initial} - V_{final}]e^{-\frac{t}{\tau}}$$

$$\text{or,} \quad v(t) = V(\infty) + [V(0) - V(\infty)]e^{-\frac{t}{\tau}}$$



- The *steady-state response* (v_{ss}) is the behaviour of the circuit a long time after an external excitation is applied.
- The complete response can be written as,

$$v(t) = V_{final} + [V_{initial} - V_{final}]e^{-\frac{t}{\tau}}$$

$$\text{or,} \quad v(t) = V(\infty) + [V(0) - V(\infty)]e^{-\frac{t}{\tau}}$$

Definition of τ

$$v(t) = V_{final} + [V_{initial} - V_{final}]e^{-\frac{t}{\tau}}$$

At $t = \tau$,

$$v(t) = V_{final} + [V_{initial} - V_{final}]e^{-1}$$

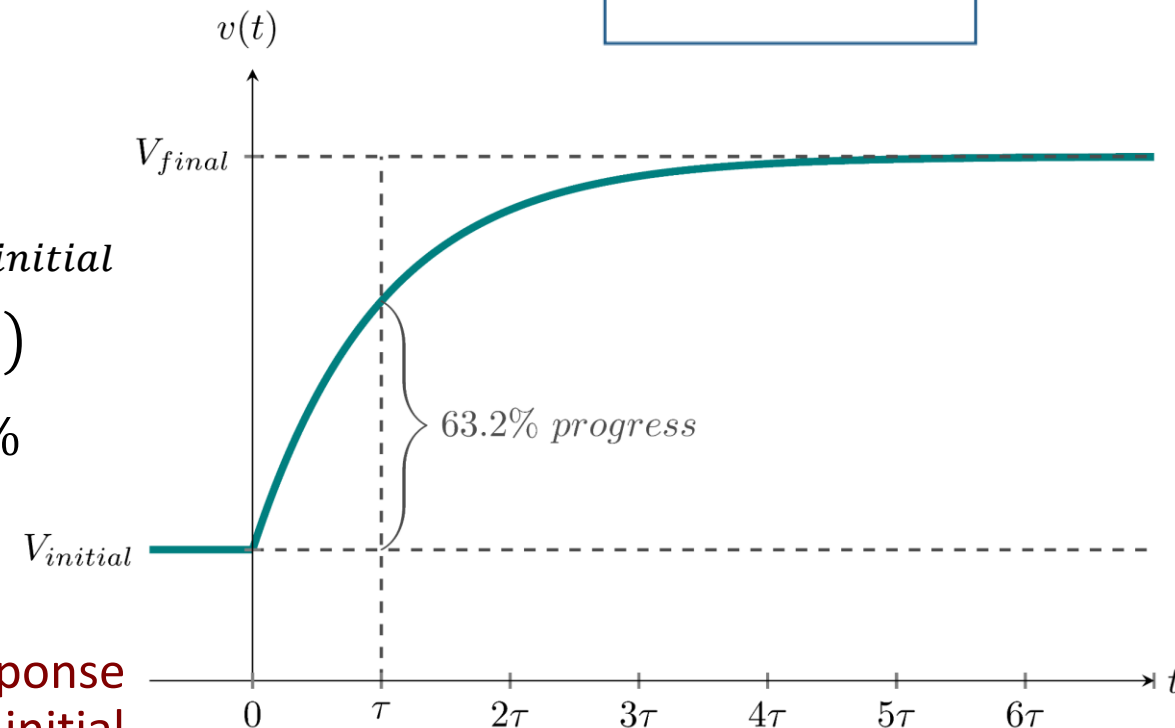
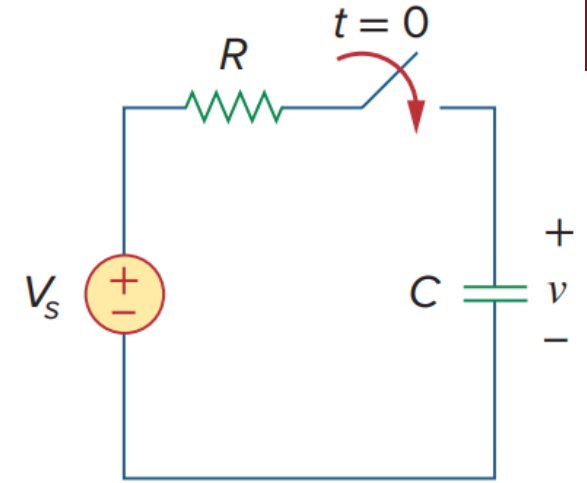
$$\Rightarrow v(t) = V_{final}(1 - 1/e) + V_{initial}(1/e)$$

$$\Rightarrow v(t) = V_{final}(1 - 1/e) - V_{initial}(1 - 1/e) + V_{initial}$$

$$\Rightarrow v(t) = V_{initial} + [V_{final} - V_{initial}](1 - 1/e)$$

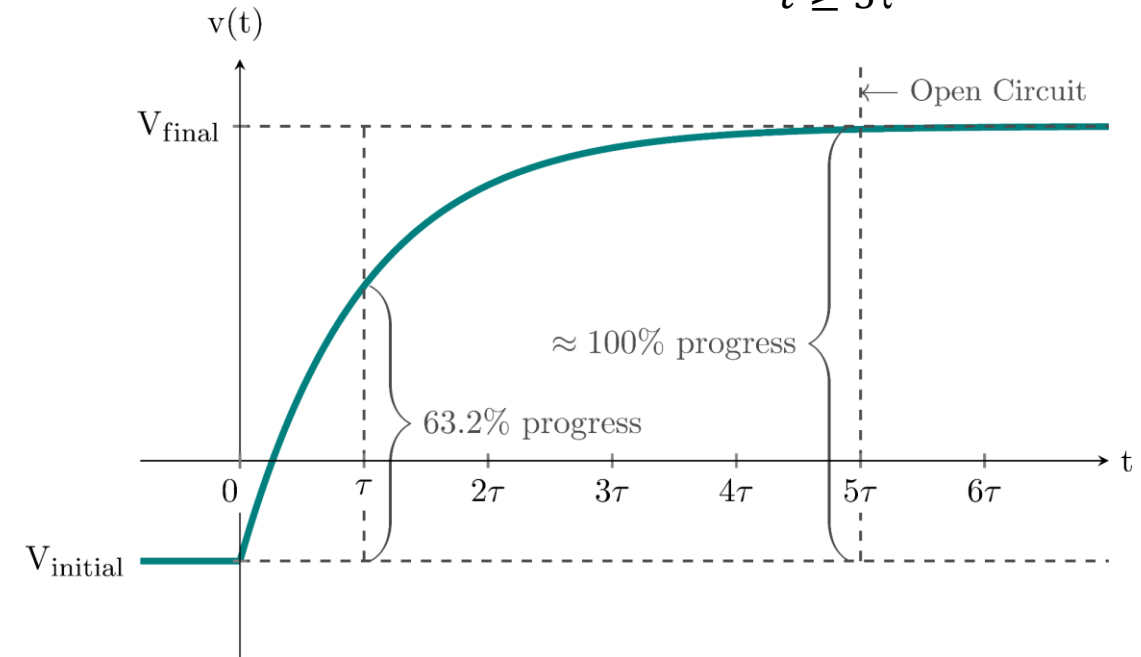
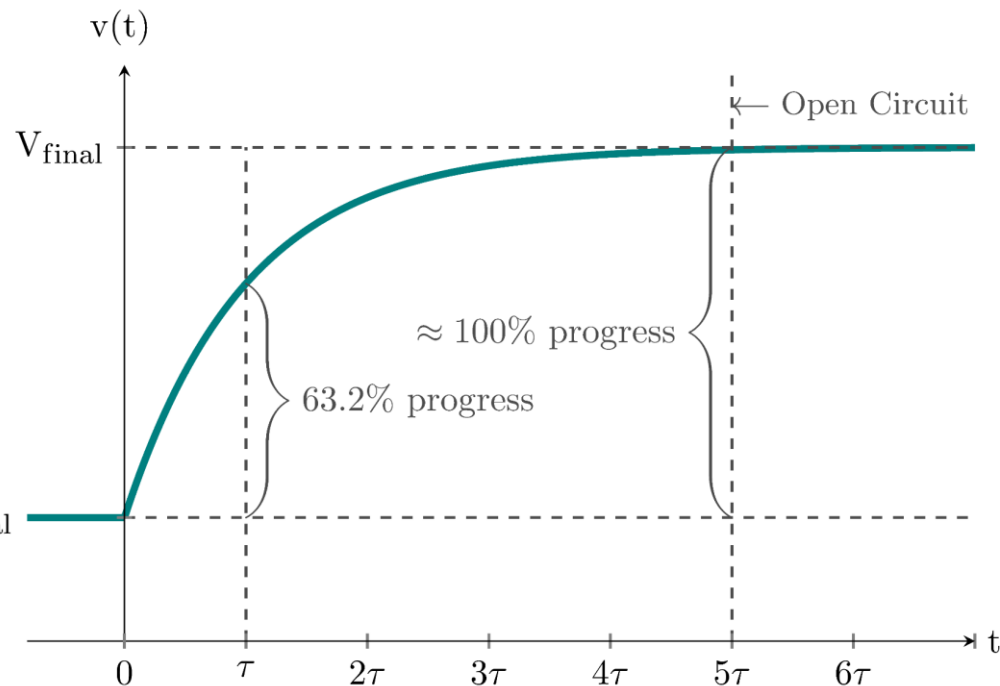
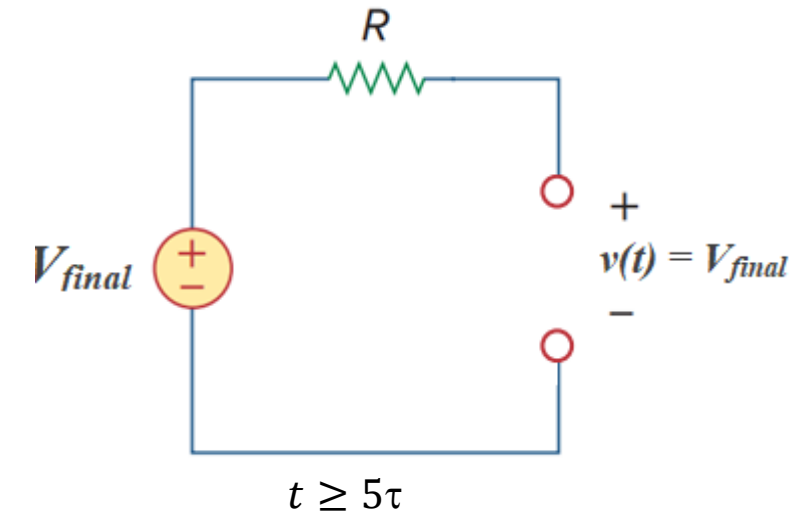
$$\Rightarrow v(t) = V_{initial} + [V_{final} - V_{initial}] \times 63.2\%$$

- We can define the time constant in this way,
- The **time constant** is the time required for the response to progress 63.2% towards V_{final} from an initial response $V_{initial}$.



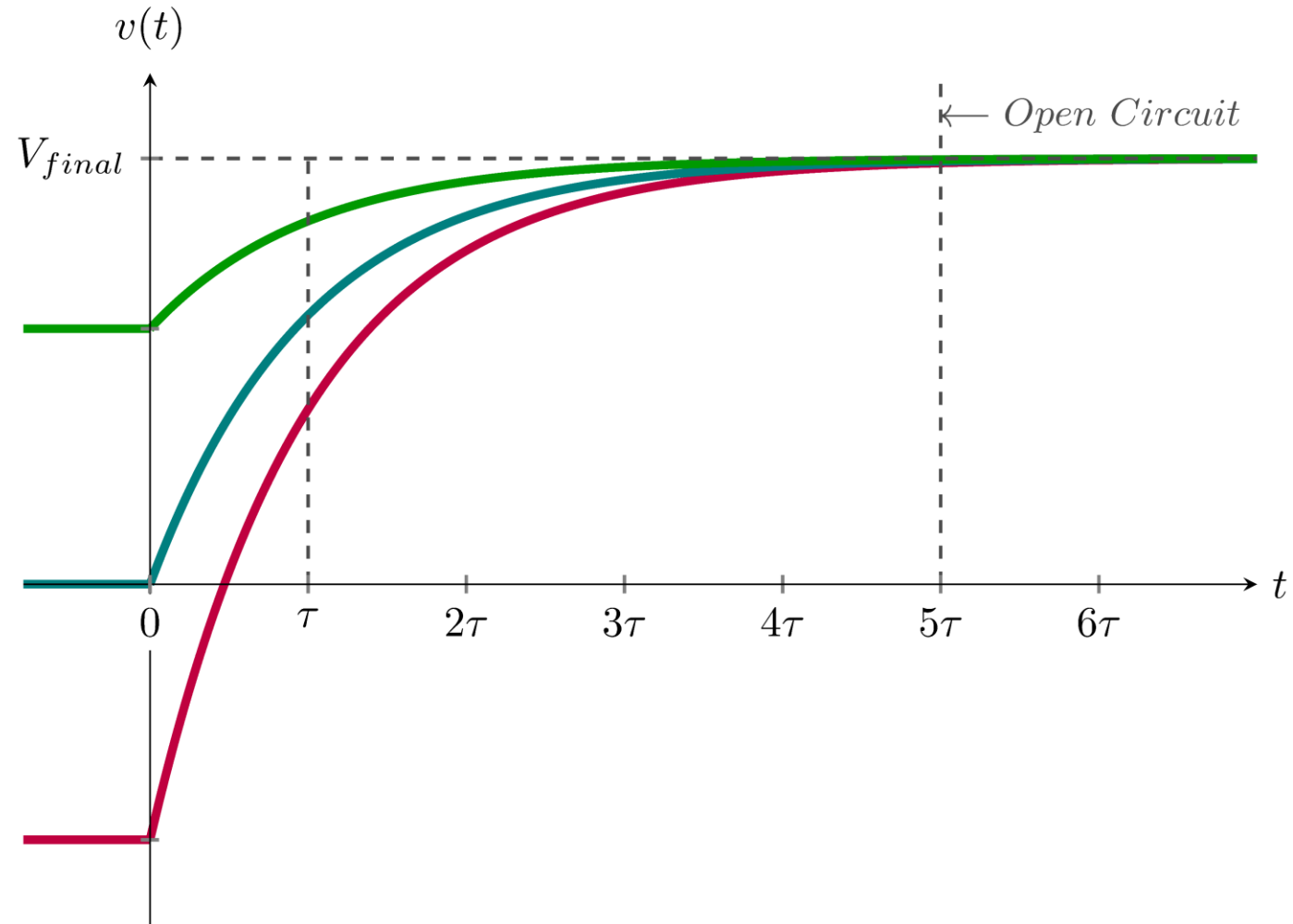
Significance of τ

- The significance of τ is that it determines how fast or slow a capacitor will be charged or discharged. Mathematically, a capacitor voltage reaches the final voltage approximately after 5 times the Time Constant (τ). The capacitor is fully charged and acts as open circuit from 5τ time onward. So, when designing circuits, the charging time of a capacitor under the application of a certain dc supply can be set by choosing R_{Th} .



τ 's independency from initial condition

- The time constant does not depend on the initial voltage or initial charge of the capacitor. For a given circuit, that is, for a fixed R_{Th} and C , the time needed for the capacitor voltage to rise to its final value is the same whether or not the capacitor is initially charged.



Source-Free RC circuit

- A *source-free RC circuit* occurs when its dc source is suddenly disconnected. The energy already stored in the capacitor is released to the resistors.
- ⇒ Assume that a capacitor is charged to V_0 and then it is connected to a resistor as shown. The capacitor starts to discharge the stored energy to the resistor.

⇒ Initially stored charge, $w(0) = \frac{1}{2}CV_0^2$

⇒ From the figure using KCL, $i_C + i_R = 0$

⇒ $C \frac{dv}{dt} + \frac{v}{R} = 0$

⇒ $\frac{dv}{dt} + \frac{v}{RC} = 0$

⇒ $\frac{dv}{v} = -\frac{1}{RC} dt$

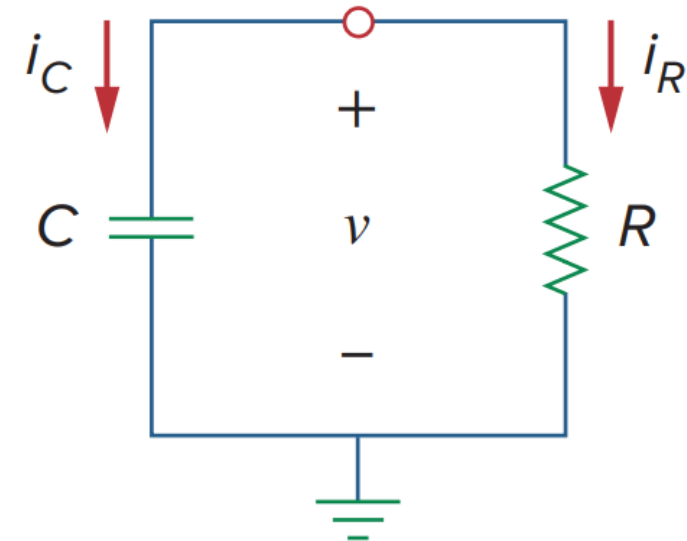
⇒ $\ln v = -\frac{t}{RC} + \ln A$

⇒ $\ln \frac{v}{A} = -\frac{t}{RC}$

⇒ $v = Ae^{-\frac{t}{RC}}$

Integrating both sides,

At $t = 0$, $v(0) = A = V_0$. So, $v(t) = V_0 e^{-\frac{t}{RC}}$



Time Constant

$$v(t) = V_0 e^{-\frac{t}{RC}}$$

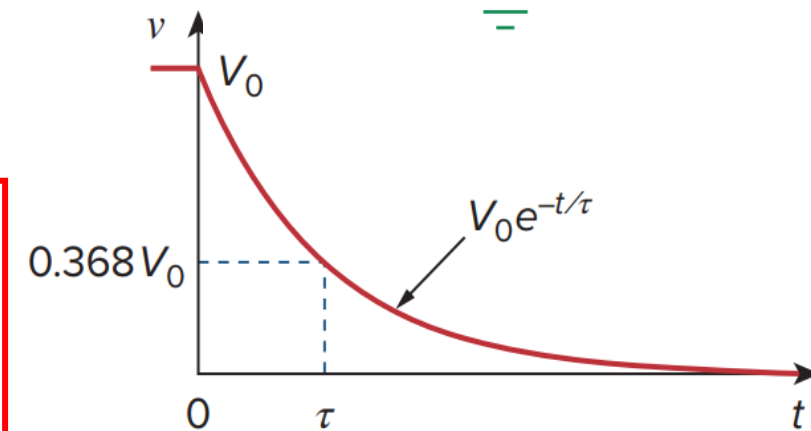
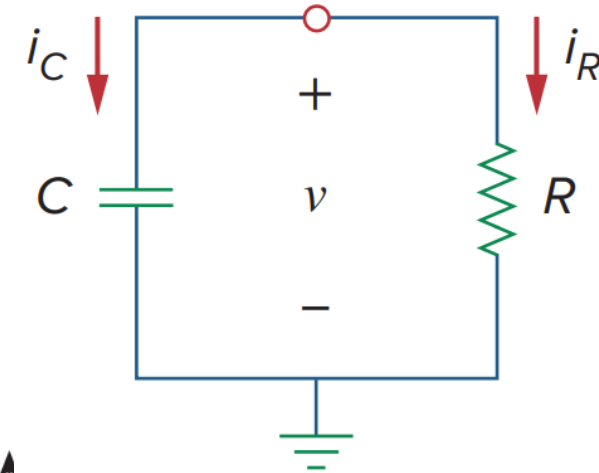
- This shows that the voltage response of the RC circuit is an exponential decay of the initial voltage. It is called the *natural response* of the circuit.

$$\Rightarrow v(t) = V_0 e^{-\frac{t}{\tau}}$$

- where $\tau = RC$ is the time constant (unit in sec).
- Notice that, we write $\tau = RC$ for the circuit consisting of only a resistor R in series with the capacitor. As we know, all the linear two terminal circuits can be reduced to this form by Thevenin's Theorem, so the resistor R is actually the Thevenin Resistance R_{Th} . Therefore,

$$\tau = R_{Th}C$$

No
Need



Definition of τ

$$v(t) = V_0 e^{-\frac{t}{\tau}}$$

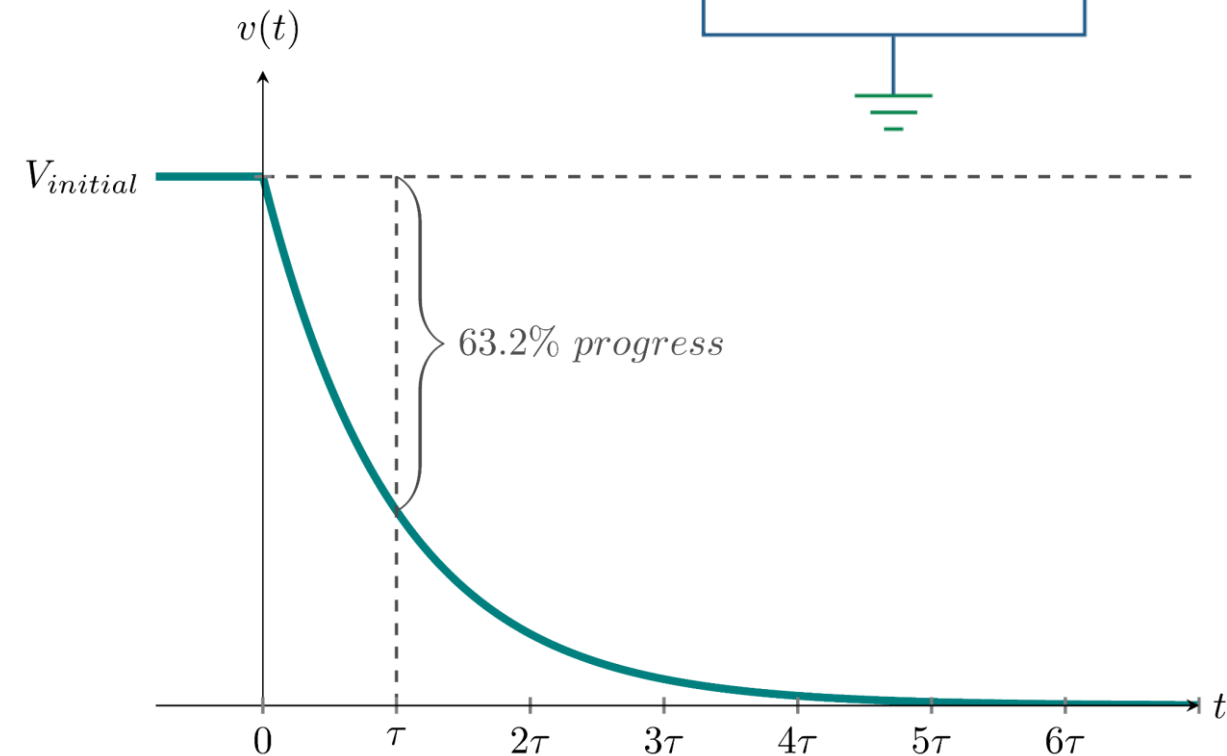
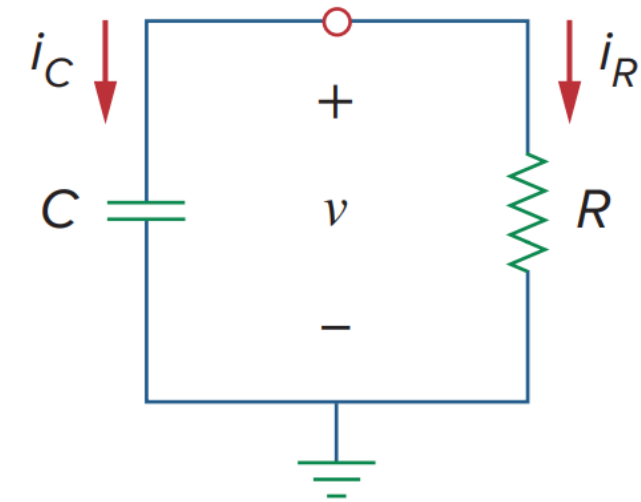
At $t = \tau$,

$$v(t) = V_0 e^{-1}$$

$$\Rightarrow v(t) = 0.368 \times V_0$$

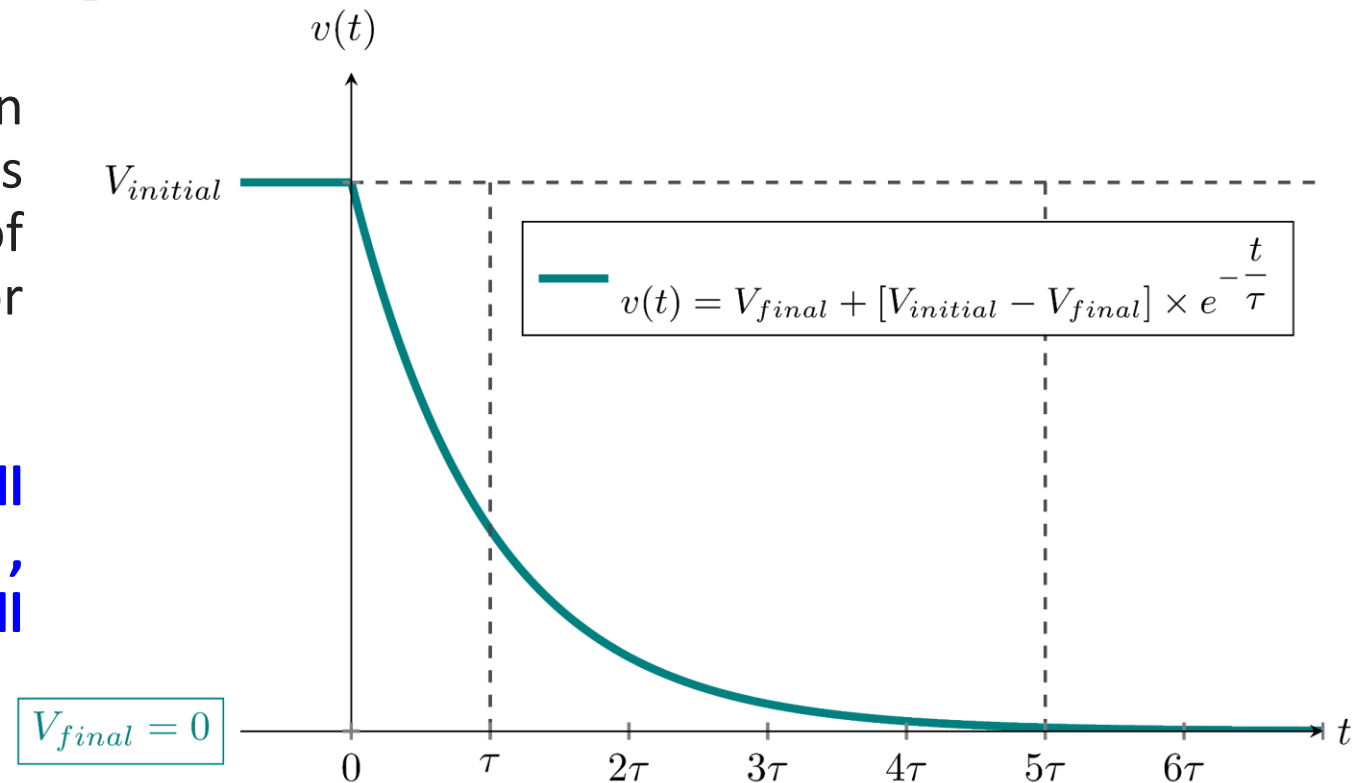
$$\Rightarrow v(t) = V_{\text{initial}} - 0.632 \times [V_{\text{initial}} - 0]$$

- We can define the time constant in the same way as we did in the case of the [step response](#):
- The *time constant* is the time required for the response to progress 63.2% towards a final voltage from an initial response V_{initial} or V_0 .



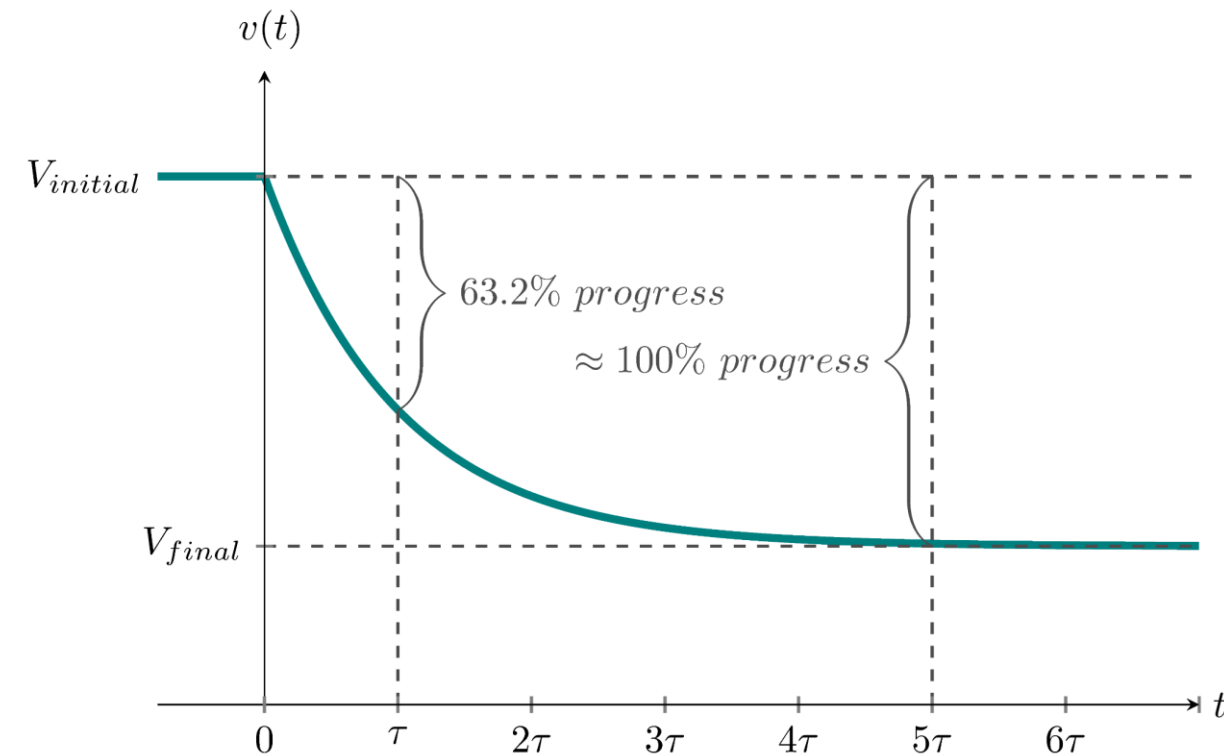
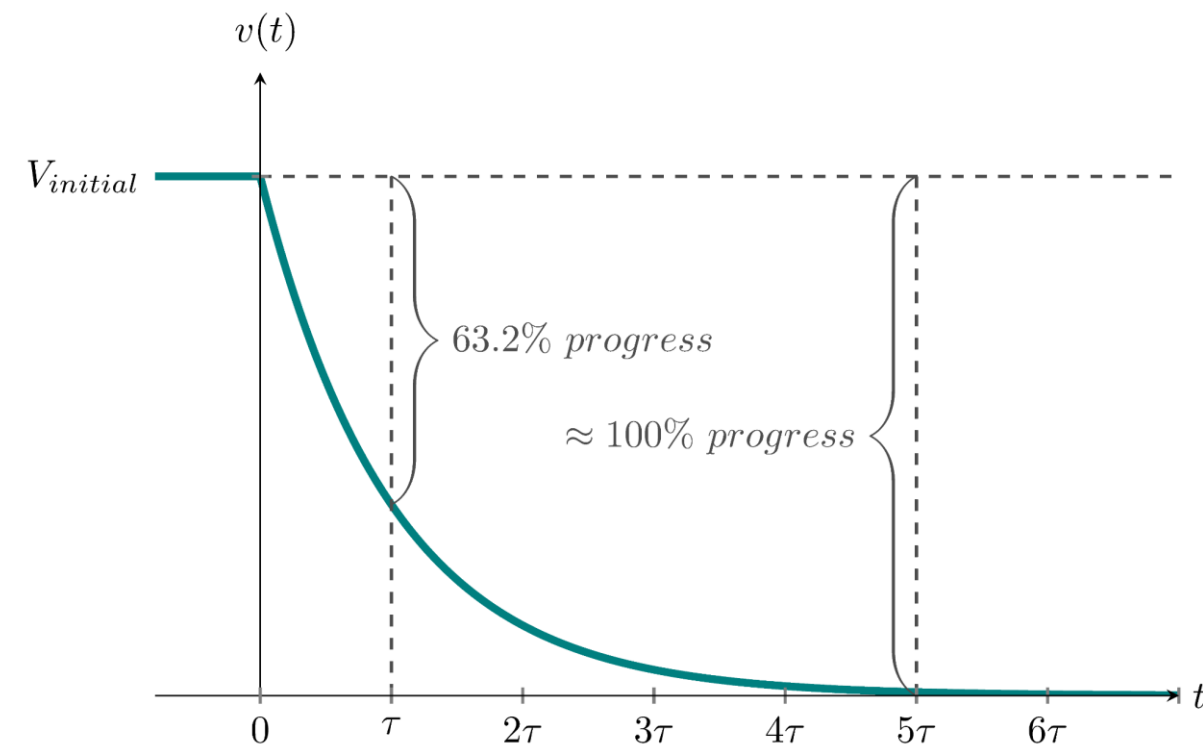
General equation

- It is interesting to interpret this way that the source-free RC circuit is a condition where $v_{final} = v(\infty) = 0$ in the step response $\left[v(t) = V(\infty) + [V(0) - V(\infty)]e^{-\frac{t}{\tau}} \right]$ or $\left[v(t) = V_{final} + [V_{initial} - V_{final}]e^{-\frac{t}{\tau}} \right]$ equation we derived earlier.
- We can in fact use this equation in general to find a capacitor's voltage response irrespective of the capacitor getting charged or discharged or both.
- A capacitor's energy will increase if $|V(\infty)| > |V(0)|$, otherwise the energy will decrease.**



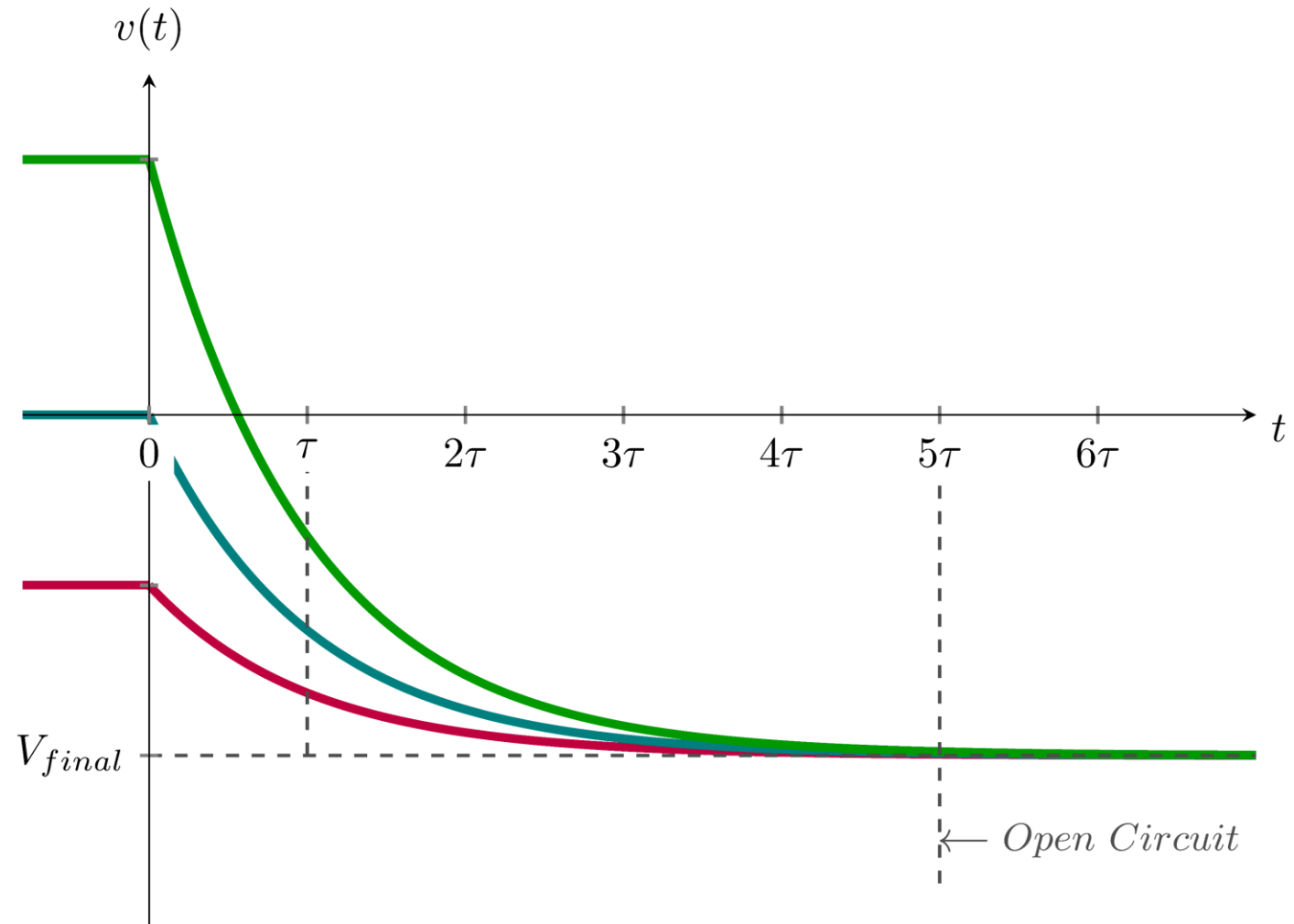
Significance of τ

- When a capacitor's initial voltage ($V_{initial}$) falls to zero ($V_{final} = 0$) (if no independent sources are present in the circuit to charge it back), it can be shown mathematically that it takes approximately 5τ time to reach the final voltage. In fact, irrespective of the values of $V_{initial}$ and V_{final} , a capacitor will always take $\approx 5\tau$ time to reach the final voltage.



τ 's independency from initial condition

- The time constant does not depend on the initial voltage or initial charge of the capacitor. For a given circuit, that is, for a fixed R_{Th} and C , the time needed for the capacitor voltage to fall to its final voltage is the same whether or not the capacitor is initially charged.



$$v(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$$

Procedure

Determine the initial voltage of the capacitor $V_{initial}$ or $V(0)$

Consider only the active[‡] portion of the circuit before switching. For example, if switching occurs at $t = 0$, consider the circuit for $t < 0$.

If the circuit includes any dc source (current or voltage), open the capacitor and determine the voltage at the open terminal. This is the $V(0)$. $V(0) = 0$ if there is no independent source in the circuit.

Determine the final voltage of the capacitor V_{final} or $V(\infty)$

Now consider the active[‡] portion of the circuit after switching. For example, for $t > 0$.

Repeat the step. This time, the voltage across the capacitor is $V(\infty)$. Circuits with $V(\infty) = 0$ are called source free.

Determine the time constant (τ)

Again, only consider the active[‡] portion after switching. For example, for $t > 0$.

Determine the Thevenin resistance (R_{Th}) as seen from the capacitor terminals

$$\tau = R_{Th}C$$

Determine $v(t)$

Plug in $V(0)$, $V(\infty)$, and τ into the equation for $v(t)$

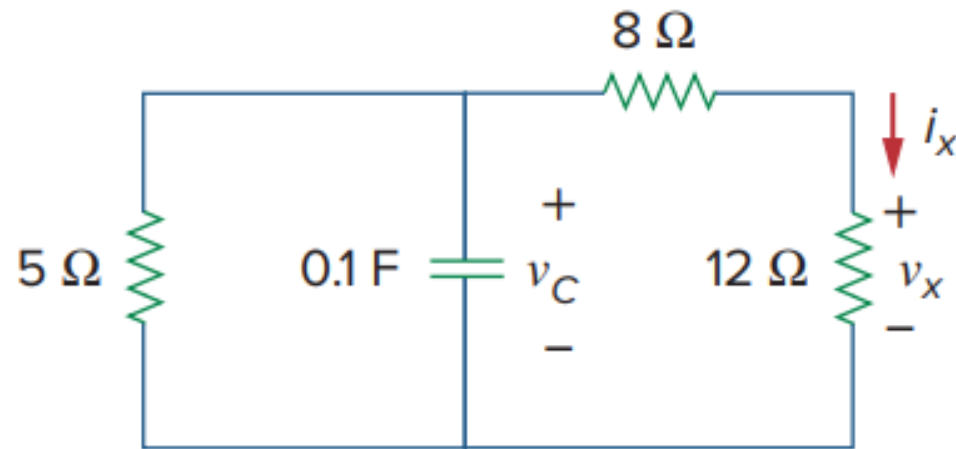
Determine any other voltages or currents in the circuit using $v(t)$ and the circuit laws.

[‡] active portion of the circuit excludes everything that has no influence on the capacitor



Example 1

- Let $V_C(0) = 15\text{ V}$, Determine v_C , v_x , and i_x for $t > 0$.



Solution

The equivalent resistance as seen from the capacitor terminal is,

$$R_{eq} = (8 + 12) \parallel 5 = 4\ \Omega$$

Time constant, $\tau = R_{eq}C = 4 \times 0.1 = 0.4\text{ s}$

Thus, for a source-free RC circuit, $V(\infty) = 0$. So,

$$v_C(t) = V(0)e^{-\frac{t}{\tau}} = 15e^{-2.5t}\text{ (V)}$$

The voltage v_x can be found by simple voltage division.

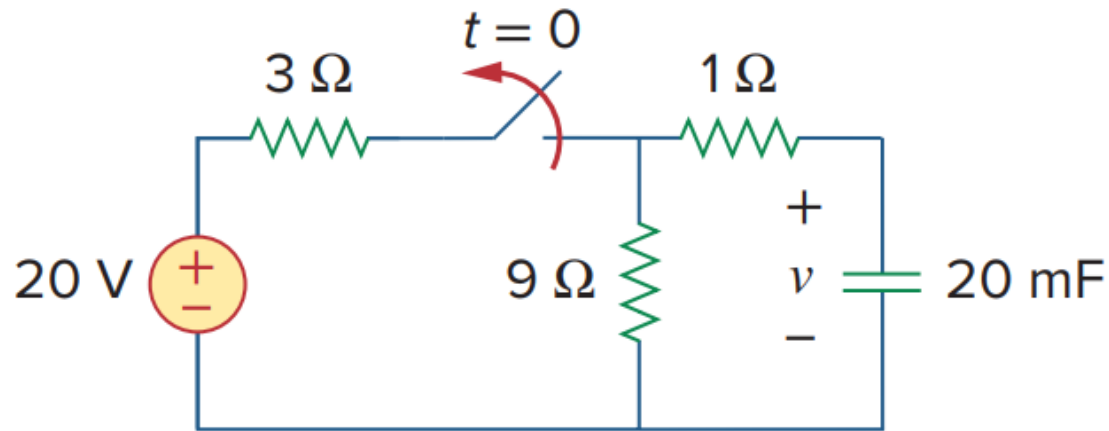
$$v_x(t) = \frac{12}{12 + 8} \times v_C(t) = 9e^{-2.5t}\text{ (V)}$$

According to the Ohm's law,

$$i_x = \frac{v_x}{12} = \frac{9e^{-2.5t}}{12} = 0.75e^{-2.5t}\text{ (A)}$$

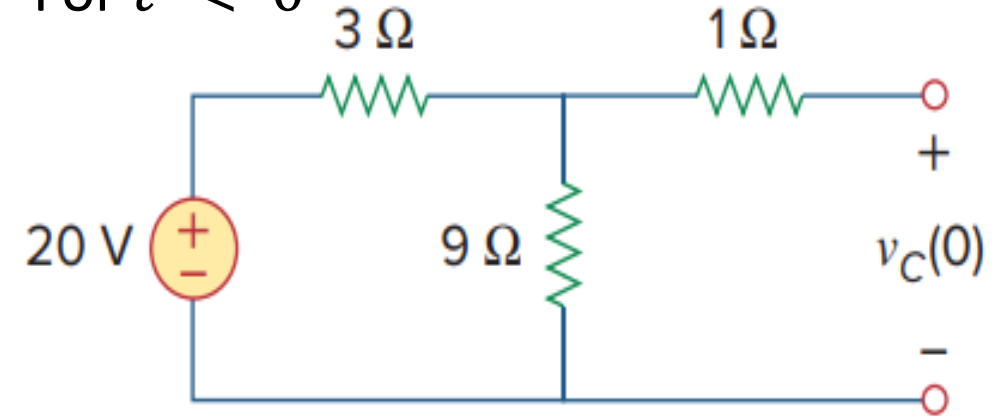
Example 2: $t < 0$

- The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t > 0$. Calculate the initial energy stored in the capacitor.



For $t < 0$

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For $t < 0$, the switch is closed. With the capacitor open at dc, the circuit transforms into the one shown above.

No current flows through the 1Ω . So, the voltage across the 9Ω is the $v_C(t)$ for $t < 0$,

$$v_C(t) = \frac{9}{9+3} \times 20 = 15\text{ V}, \quad t < 0$$

Since the voltage across the capacitor cannot change instantaneously,

$$v_C(0) = v_C(0^-) = 15\text{ V}$$



Example 2: $t > 0$

For $t > 0$, the switch is open. The circuit transforms into the one shown above. As there is no independent source in the circuit, $V(\infty) = 0$.

The Thevenin resistance as seen from the capacitor terminal,

$$R_{Th} = 1 + 9 = 10 \Omega$$

The time constant is,

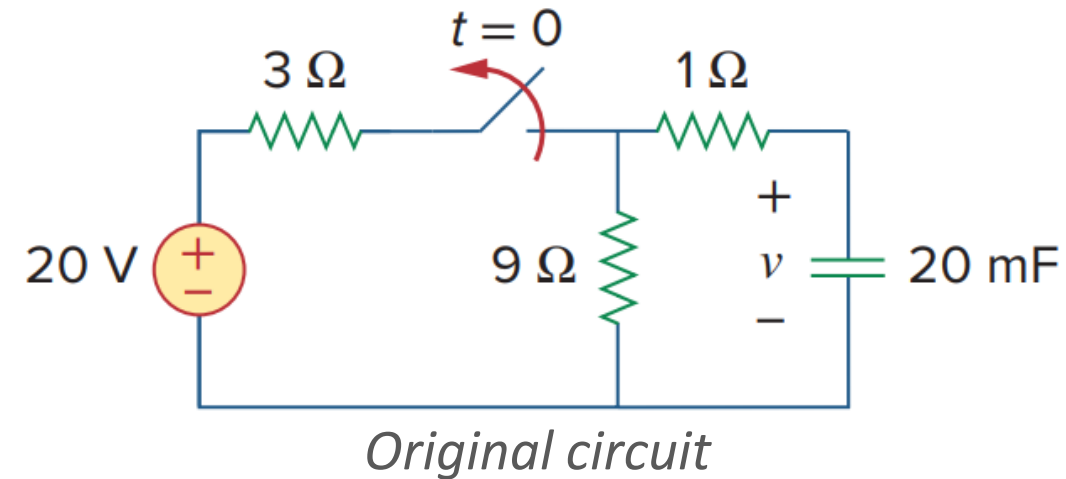
$$\tau = R_{Th}C = 10 \times 20 \times 10^{-3} = 0.2 \text{ s}$$

So, the voltage across the capacitor for $t > 0$ is,

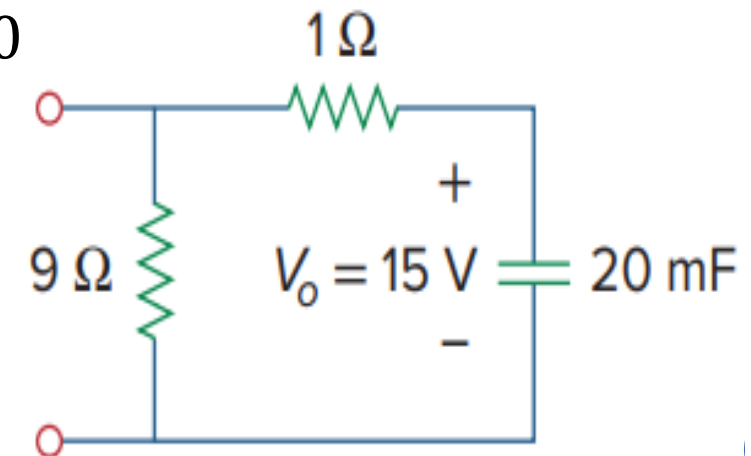
$$\begin{aligned} v_C(t) &= V(0)e^{-\frac{t}{\tau}} \\ &= 15e^{-5t} \text{ (V)} \end{aligned}$$

The initial energy stored in the capacitor is,

$$\begin{aligned} w_C(t) &= \frac{1}{2}CV(0)^2 \\ &= \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \text{ J} \end{aligned}$$

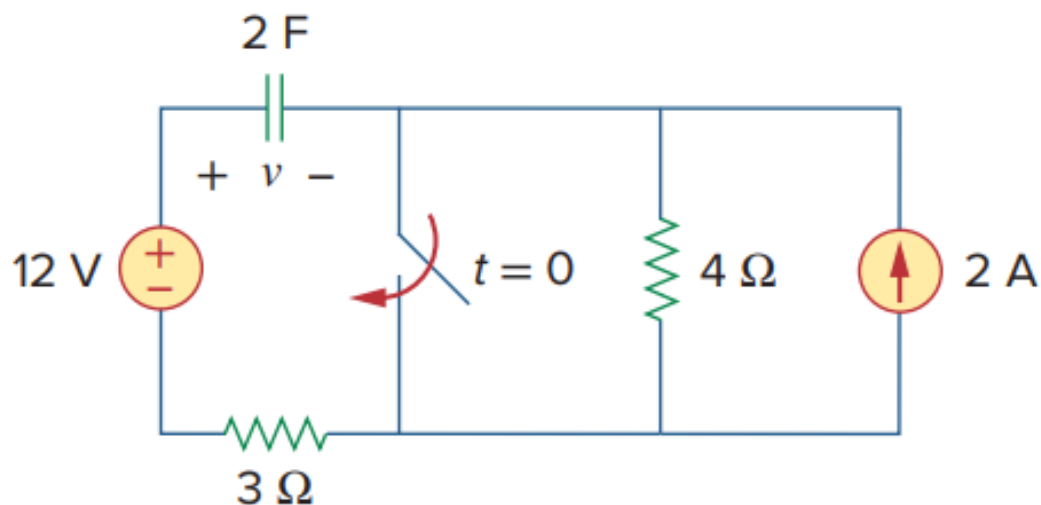


For $t > 0$

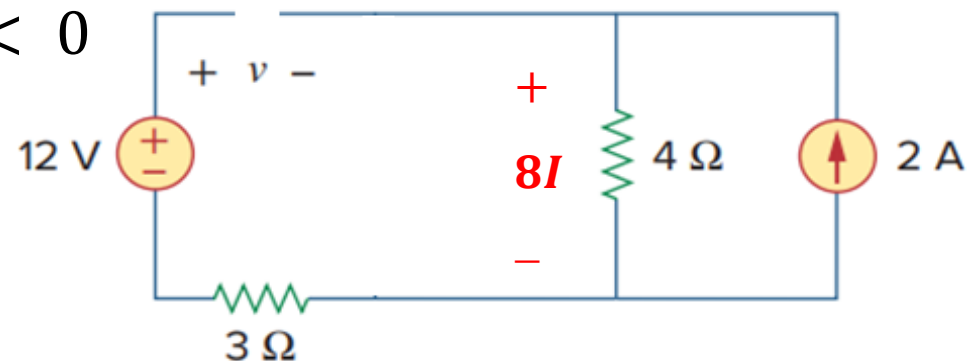


Example 3

- Calculate the capacitor voltage $v(t)$ for $t < 0$ and for $t > 0$.



For $t < 0$



For $t < 0$, the switch is open. With the capacitor open at dc, the circuit transforms into the one shown above.

The 2 A current from the current source will flow only through the 4 Ω resistance. The voltage drop across the 4 Ω resistance is, $4 \times 2 = 8 \text{ V}$.

There is no voltage drop across the 3 Ω ($i = 0$ at open circuit). So,

$$v(t) = 12 - 8 = 4 \text{ V}, \quad t < 0$$

Since the voltage across the capacitor cannot change instantaneously,

$$v(0) = v(0^-) = 4 \text{ V}$$

Example 3: $t > 0$

For $t > 0$, the switch is closed. With the capacitor again open at dc, the circuit transforms into the one shown above.

Again, there is no voltage drop across the $3\ \Omega$ ($i = 0$ at open circuit). So,

$$v(t) = 12\text{ V}, \quad t > 0$$

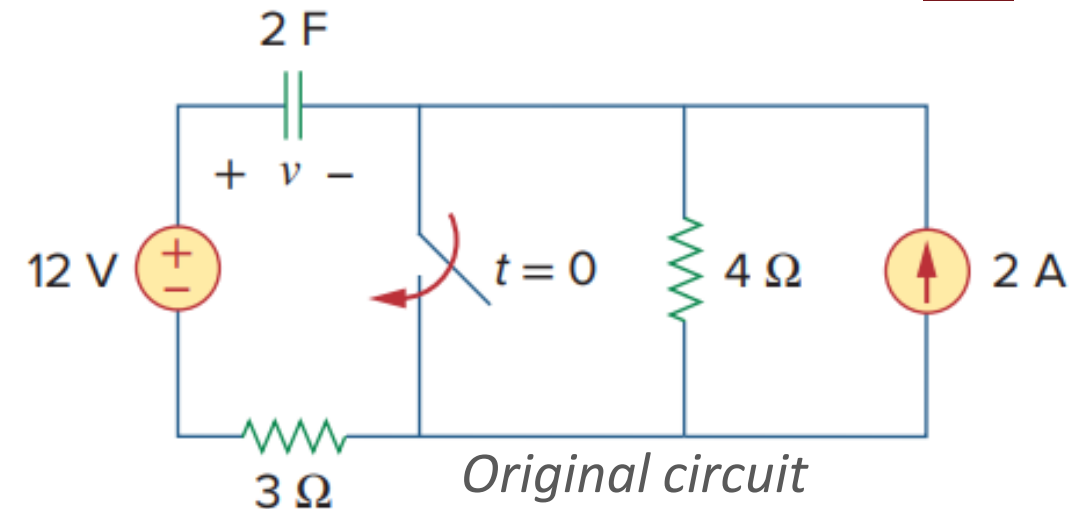
This is the steady-state voltage across the capacitor for $t > 0$.

$$v(\infty) = 12\text{ V}$$

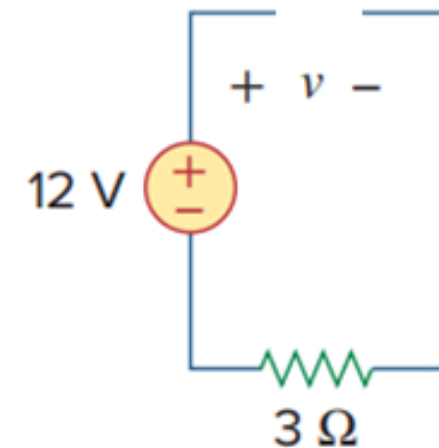
The time constant is, $\tau = R_{Th}C = 3 \times 2 = 6\text{ s}$

So,

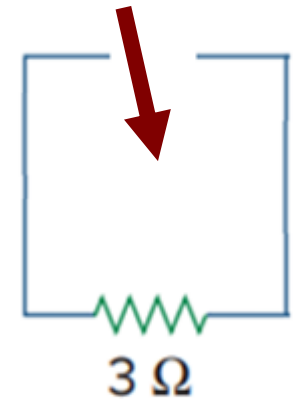
$$\begin{aligned} v(t) &= V(\infty) + [V(0) - V(\infty)]e^{-t/\tau} \\ &= 12 + [4 - 12]e^{-\frac{t}{6}} = 12 - 8e^{-\frac{t}{6}} \end{aligned}$$



For $t > 0$

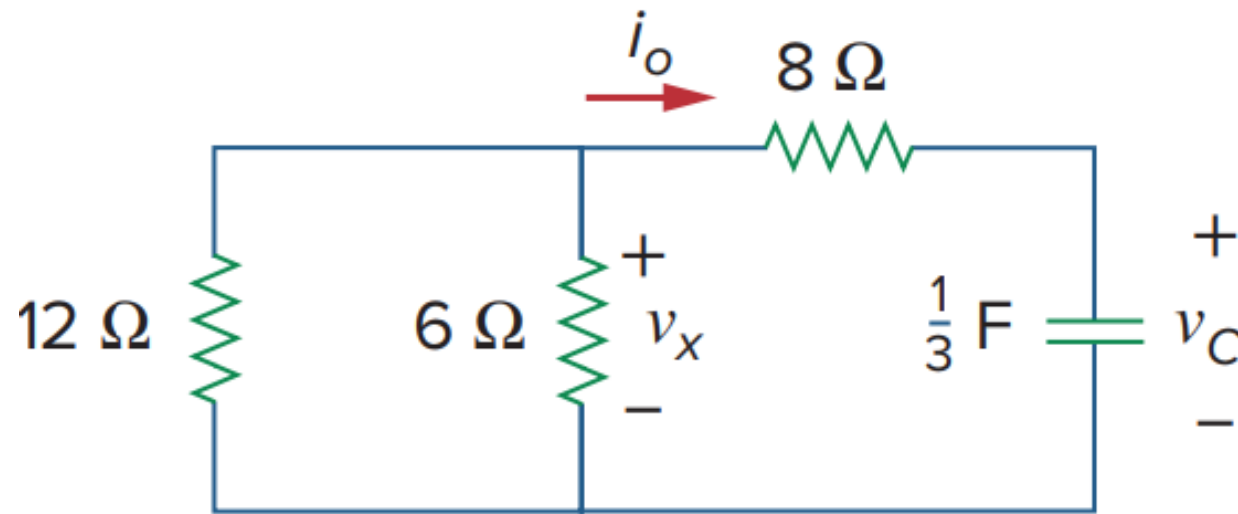


$$R_{Th} = 3\ \Omega$$



Problem 1

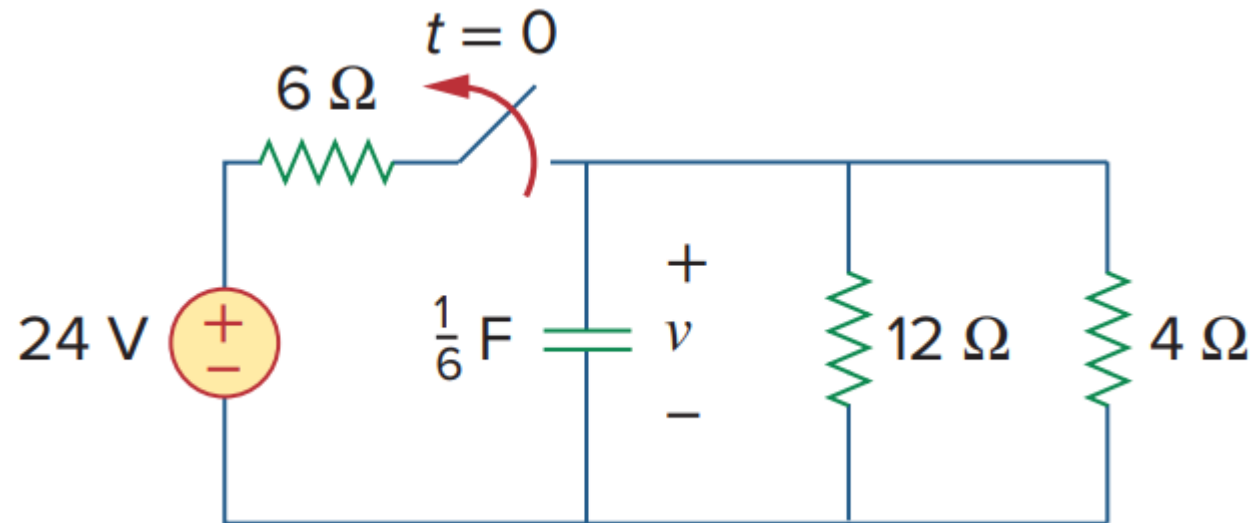
- Let $V_C(0) = 60\text{ V}$, Find v_C , v_x , and i_x for $t > 0$.



Ans: $v_C = 60e^{-0.25t}\text{ V}$; $v_x = 20e^{-0.25t}\text{ V}$; $i_x = -5e^{-0.25t}\text{ A}$

Problem 2

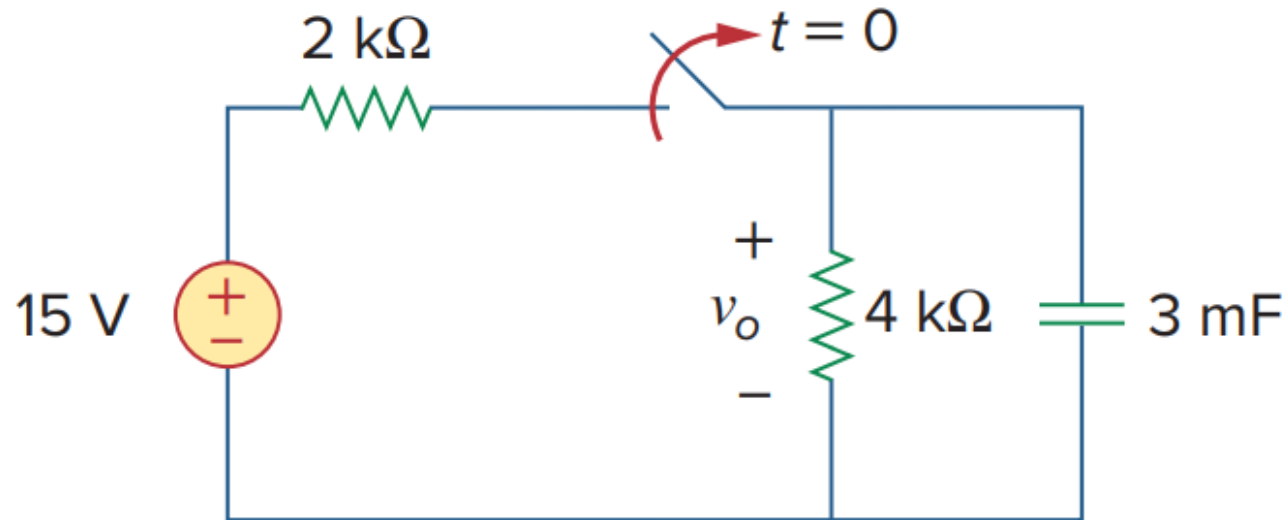
- The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t > 0$. Calculate the initial energy stored in the capacitor.



Ans: $v(t) = 8e^{-2t}\text{ V}; w_C(0) = 5.333\text{ J}$

Problem 3

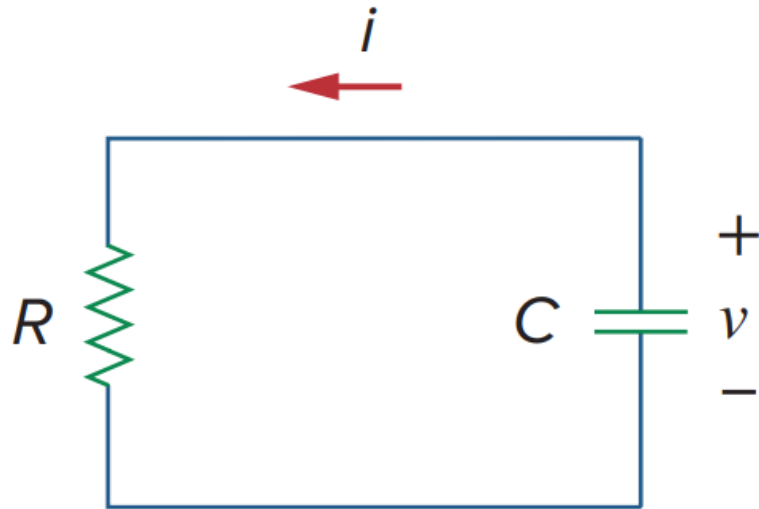
- The switch opens at $t = 0$. Find $v_o(t)$ for $t > 0$.



Ans: $v(t) = 10e^{-t/12} \text{ V}$

Problem 4

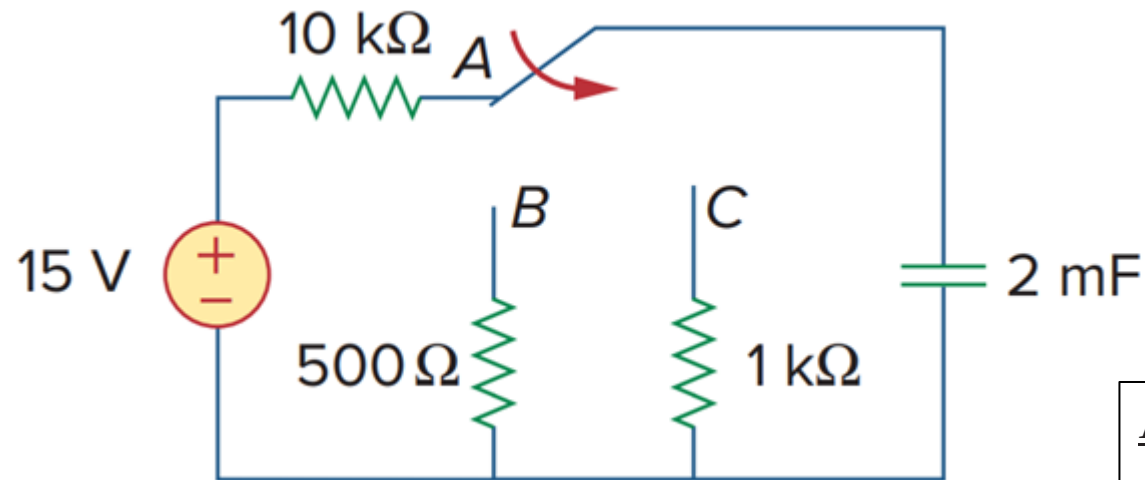
- For the circuit below, $v = 10e^{-4t} \text{ V}$ and $i = 0.2e^{-4t} \text{ A}$
- (a) Find R and C .
- (b) Determine the time constant.
- (c) Calculate the initial energy in the capacitor.
- (d) Obtain the time it takes to dissipate 50% of the initial energy.



Ans: $R = 50 \Omega$; $C = 5 \text{ mF}$; $\tau = 0.25 \text{ s}$; $w_{C(0)} = 0.25 \text{ J}$; $t = 86 \text{ ms}$

Problem 5

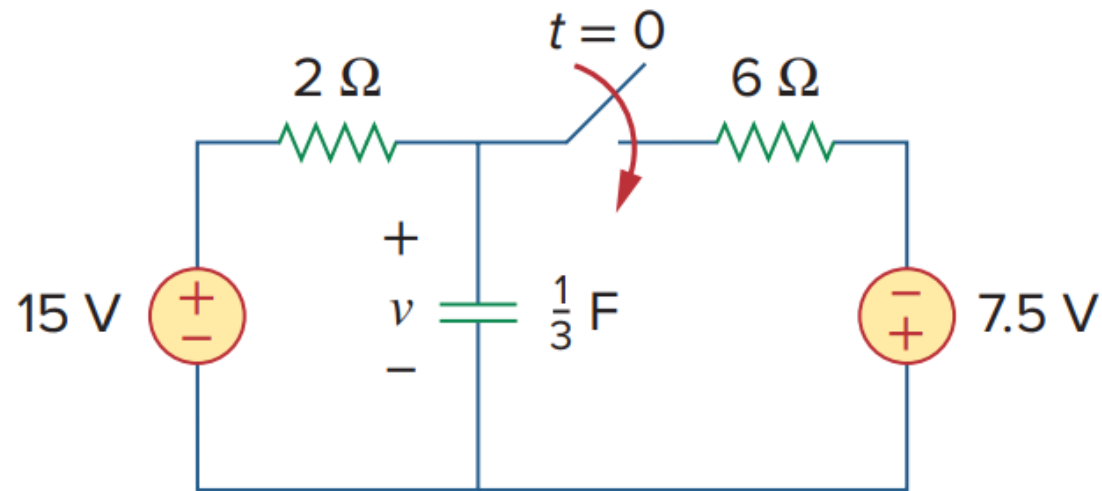
- Assume that the switch has been in position A for a long time and is moved to position B at $t = 0$. Then at $t = 1\text{s}$, the switch moves from B to C. Find $I_C(t)$ for $t > 0$.



Ans: $v(t) = 15e^{-t} \text{ V for } 0 < t < 1 \text{ sec};$
 $v(t) = 5.518e^{-(t-1)/2} \text{ V for } 1 < t < \infty \text{ sec};$

Problem 6

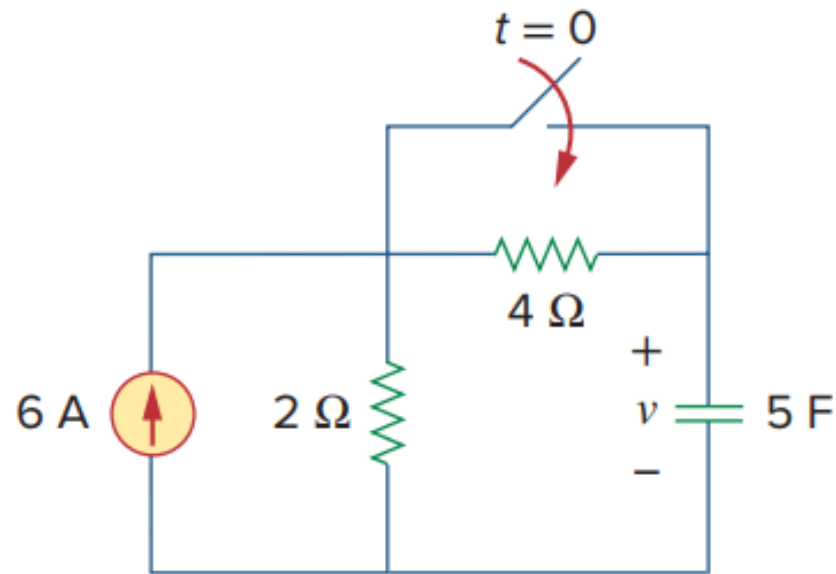
- Find $v(t)$ for $t > 0$ in the circuit shown below. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5s$.



Ans: $v_c(t) = 9.375 + 5.625e^{-2t} \text{ V for } t > 0; v_c(0.5) = 11.444 \text{ V}$

Problem 7

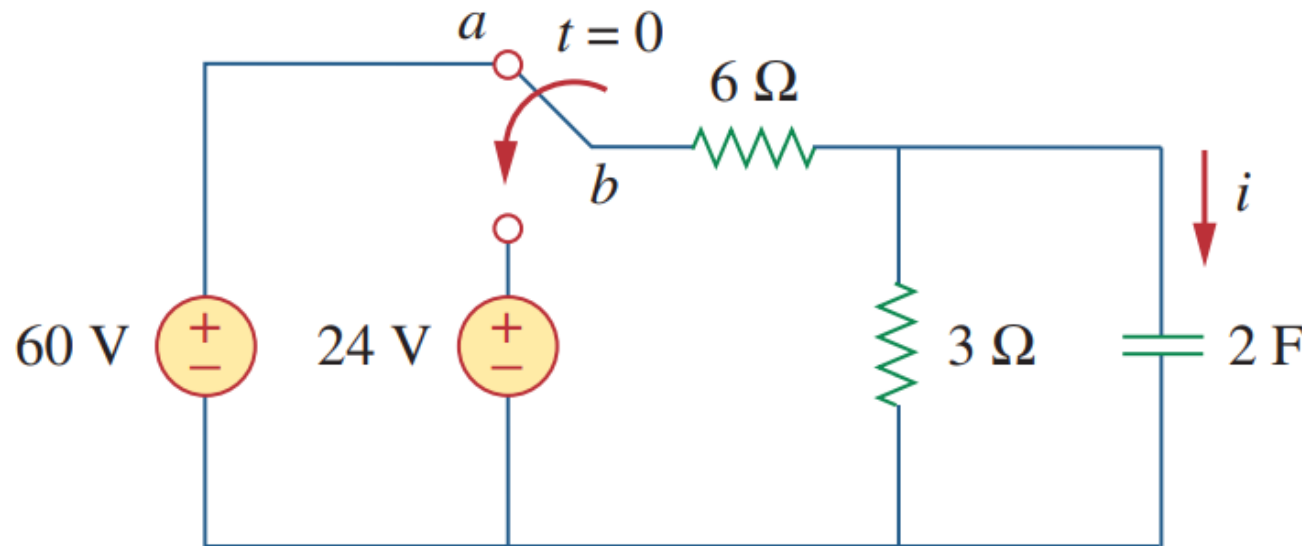
- Calculate the capacitor voltage for $t < 0$ and for $t > 0$.



Ans: $v(t) = 12\text{ V for } t < 0$; $v(t) = 12\text{ V for } t > 0$

Problem 8

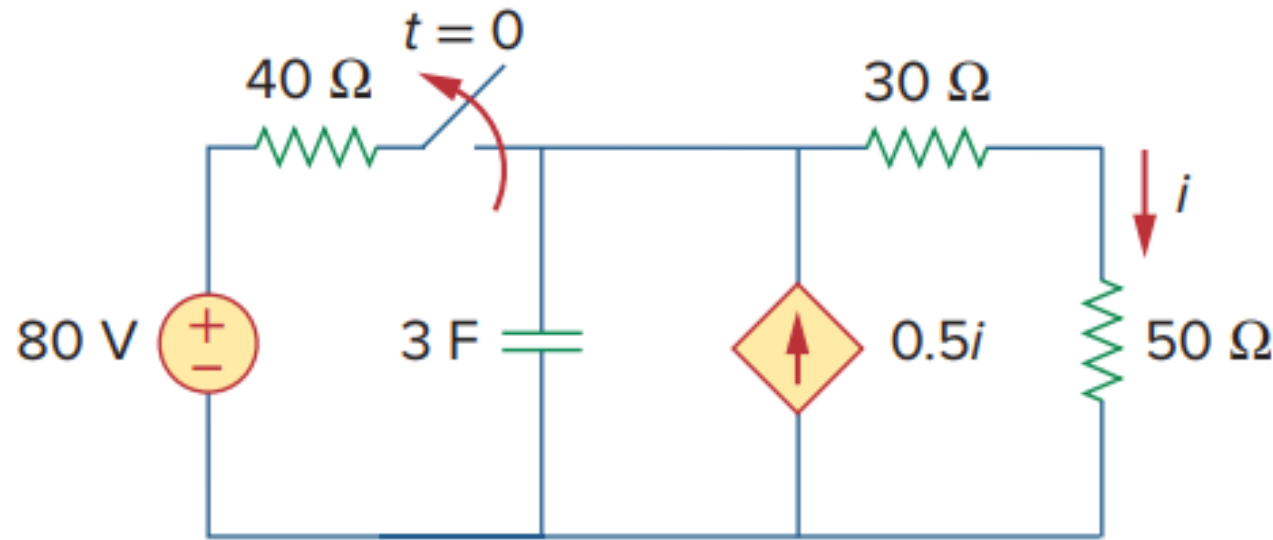
- The switch has been in position a for a long time. At $t = 0$ it moves to position b . Calculate $i(t)$ for all $t > 0$.



Ans: $i(t) = -6e^{-0.25t} \text{ A for } t > 0$

Problem 9

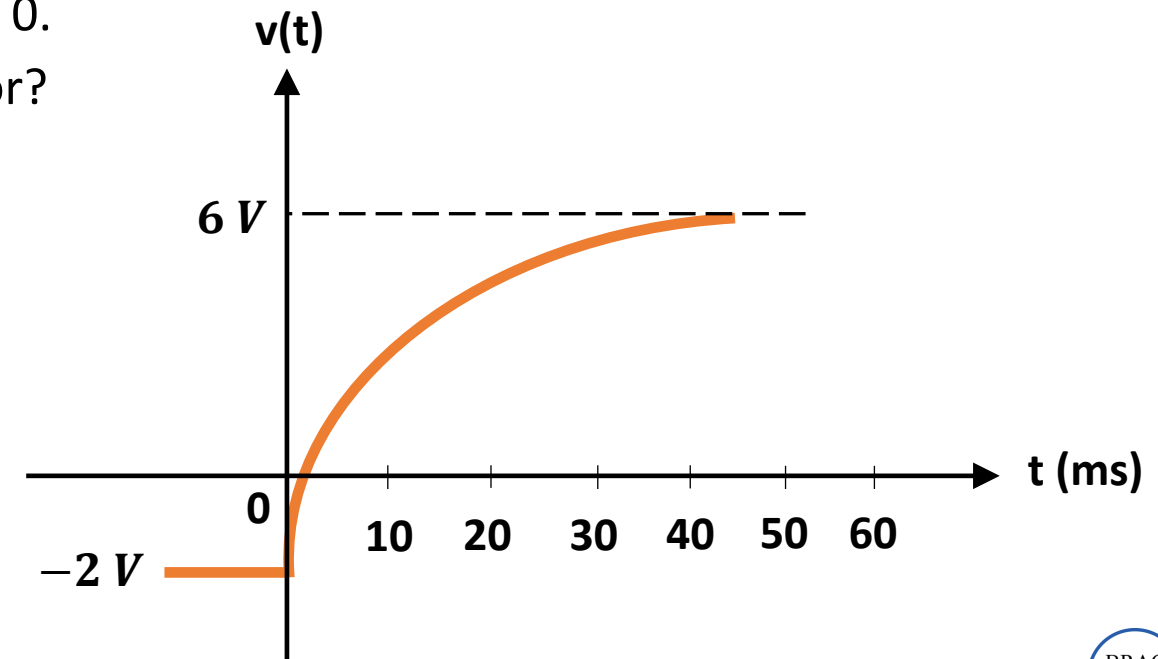
- Consider the circuit shown below. Find $i(t)$ for $t < 0$ and $t > 0$.



Ans: $i(t) = 0.8 \text{ A for } t < 0$; $i(t) = 0.8e^{-t/480} \text{ A for } t > 0$

Problem 10

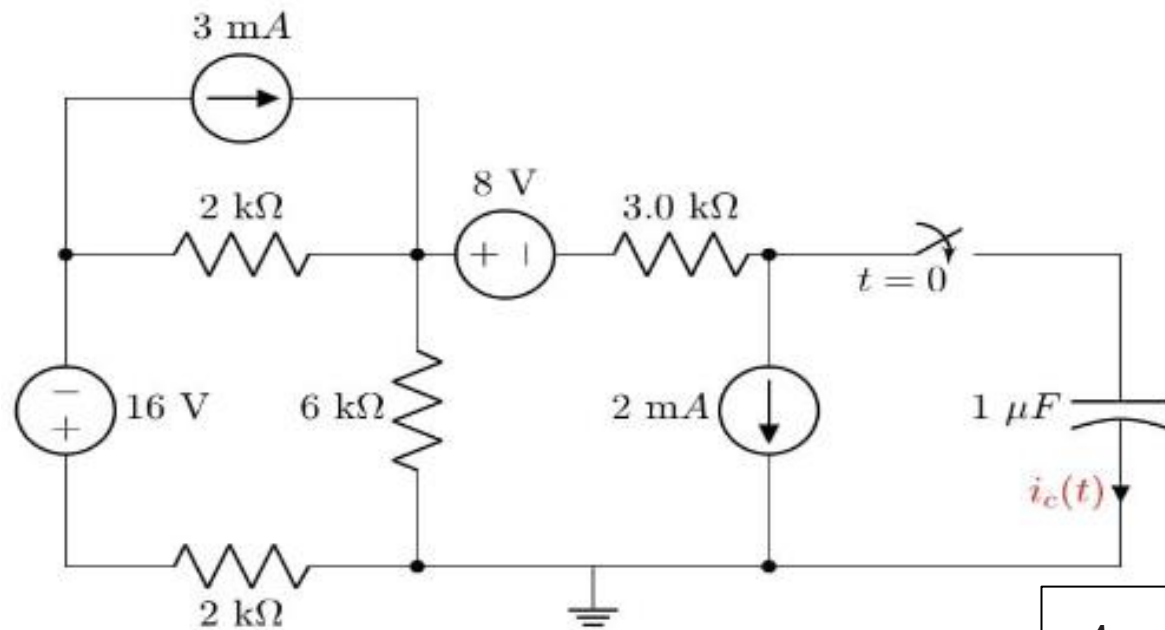
- The figure below shows the voltage response of an RC circuit to a sudden DC voltage applied through an equivalent resistance of $4\text{ k}\Omega$.
 - Define time constant.
 - Determine the approximate time constant from the figure.
 - Find the mathematical expression of $v(t)$ for $t > 0$.
 - What is the initial energy stored in the capacitor?
 - Draw the circuit diagram.



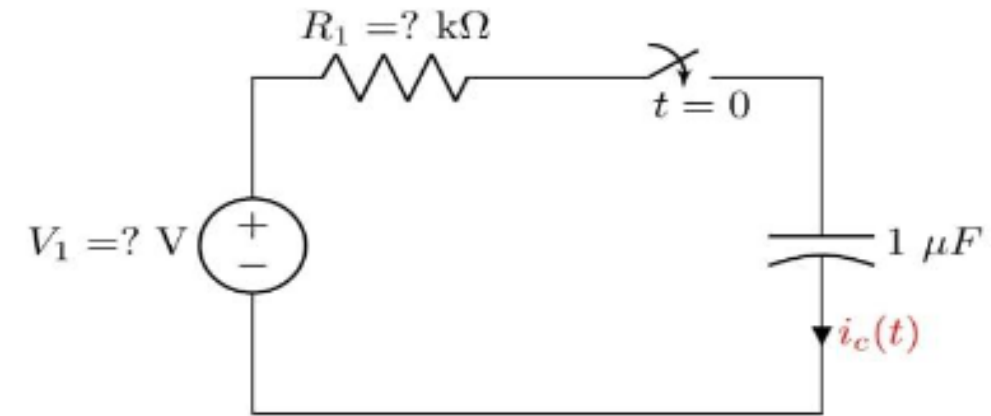
Ans: (ii) $\tau = 9\text{ ms}$; (iii) $v(t) = 6 - 8e^{-1000t/9}\text{ V for } t > 0$;
 (iv) $w = 4.5 \times 10^{-6}\text{ J}$

Problem 11

- I. Simplify the circuit 1 below so that it takes the form of the circuit 2. Determine the values of V_1 and R_1 .
- II. Perform transient analysis to determine $i_c(t)$ through the capacitor for $t > 0$.



Circuit 1

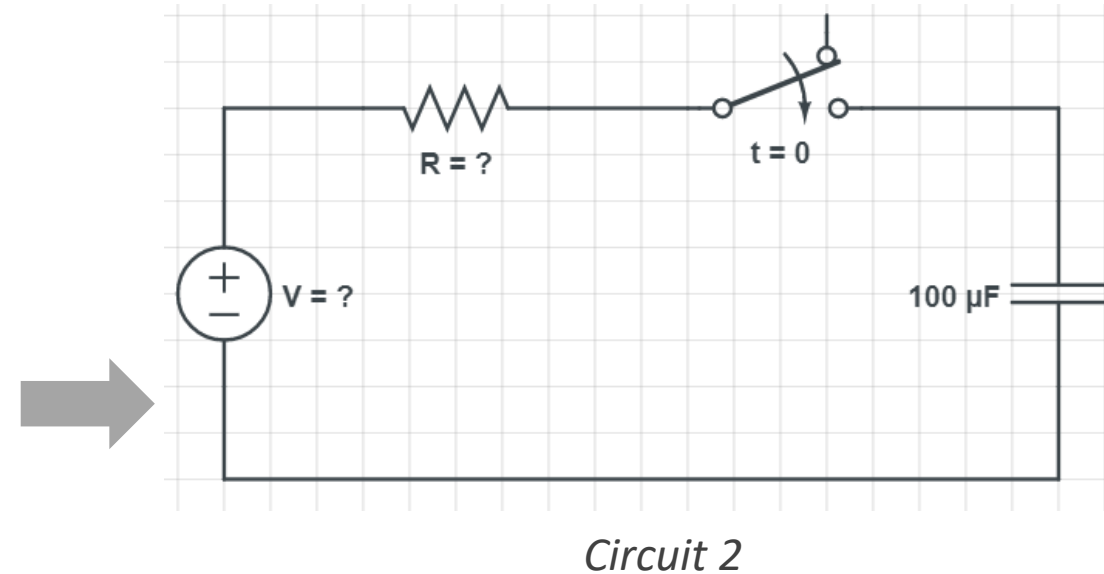
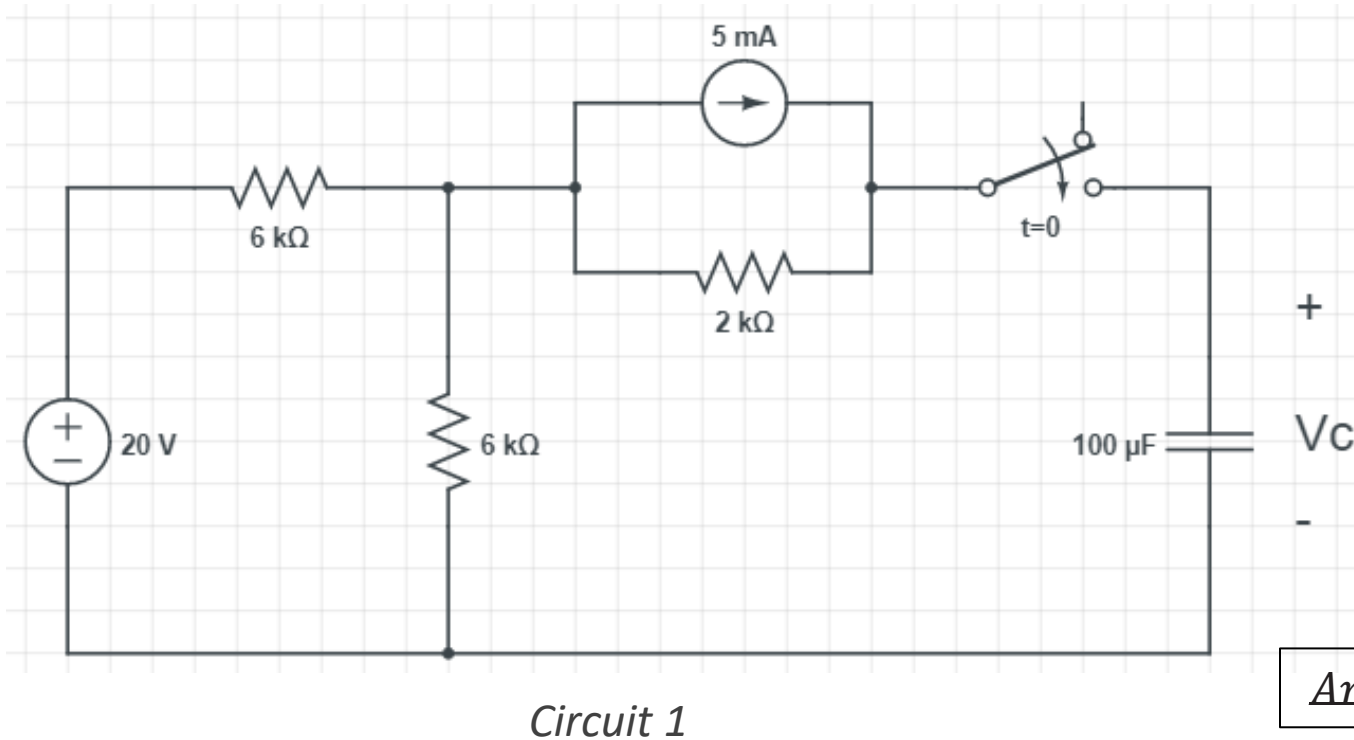


Circuit 2

Ans: $V_1 = -24.8 \text{ V}$; $R_1 = 5.4 \text{ k}\Omega$; $i_c(t) = -4.6e^{-1000t/5.4} \text{ A}$

Problem 12

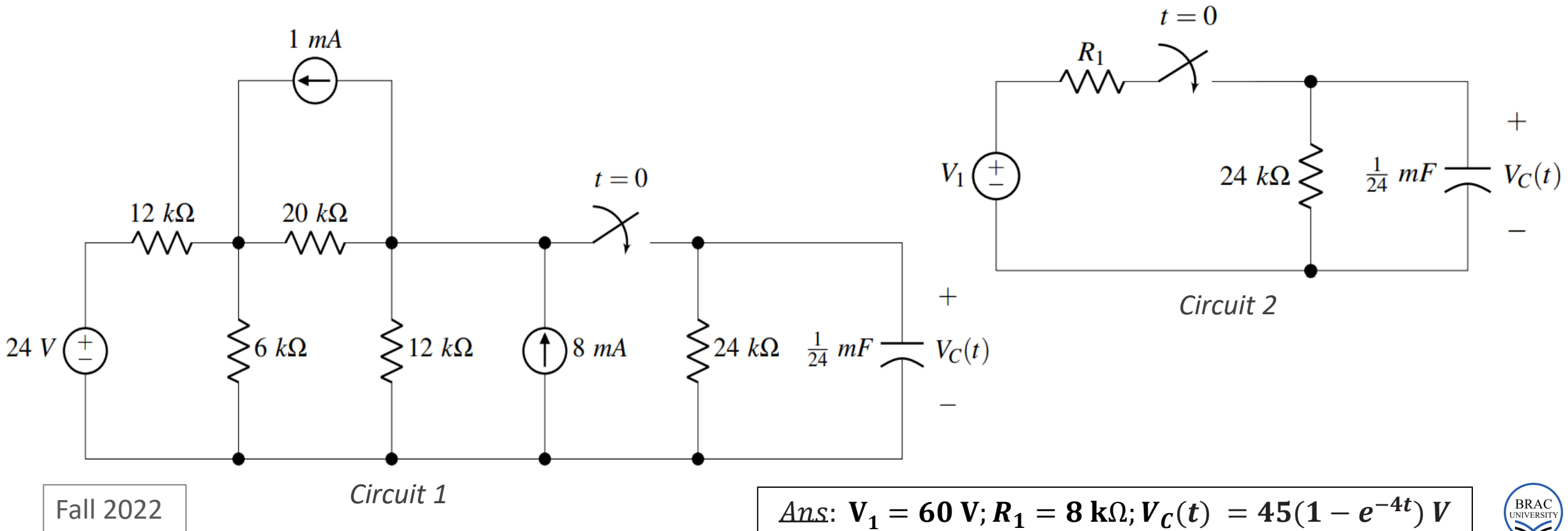
- Simplify the Circuit 1 below so that it takes the form of the Circuit 2. Determine the values of I and R .
- Perform transient analysis to determine $V_C(t)$ across the capacitor for $t > 0$.



Ans: $V = 20 \text{ V}; R = 5 \text{ k}\Omega; V_C(t) = 20(1 - e^{-2t}) \text{ V}$

Problem 13

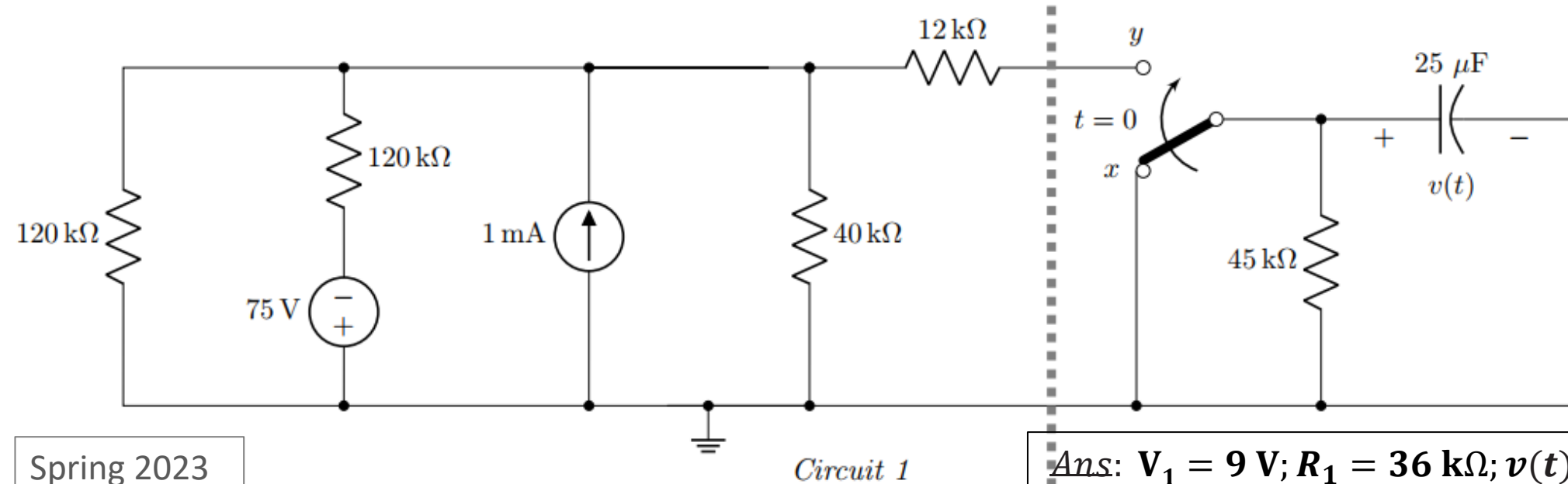
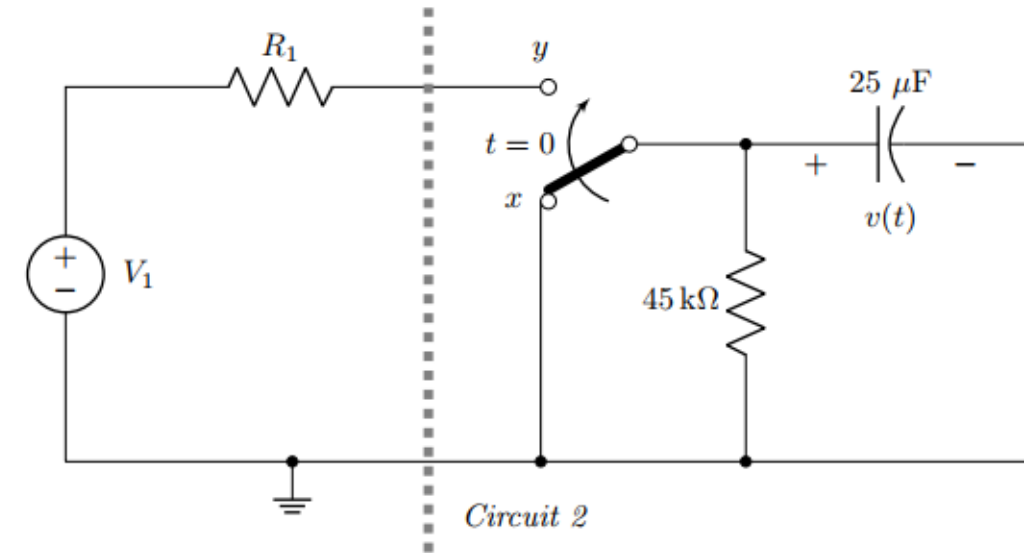
- Simplify the Circuit 1 below so that it takes the form of the Circuit 2. Determine the values of V_1 and R_1 . Perform transient analysis to determine $V_C(t)$ across the capacitor for $t > 0$.



Ans: $V_1 = 60 \text{ V}; R_1 = 8 \text{ k}\Omega; V_C(t) = 45(1 - e^{-4t}) \text{ V}$

Problem 14

- Reduce the left portion with respect to the dashed grey line of Circuit 1 so that it takes the form of Circuit 2 as shown. Write down the values of V_1 and R_1 .
- Now, analyse the Transient Behaviour of the circuit assuming that the switch moves from position x to position y at $t = 0$. Determine $v(t)$ for $t > 0$.



Ans: $V_1 = 9 \text{ V}; R_1 = 36 \text{ k}\Omega; v(t) = 5(1 - e^{-2t}) \text{ V}$